Are the phases real? Distinguishing bounded interaction groups from spatially structured drift in central Mississippi Valley decorated ceramics

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Abstract

The phase, one of archaeology's foundational units, rests on the silent premise that assemblages more similar to those nearby than to distant neighbors record a bounded society. While the validity of phases has been questioned, the rationale for assuming these units is rarely tested. Neutral transmission produces the same clustering, with no bounded groups, since drift locally concentrates similar styles. For the late pre-contact central Mississippi Valley, we test whether decorated ceramic similarity records reflect bounded interaction groups or spatially structured drift. We compare drift and bounded-groups processes on the region's geographic data, calibrated to the record's small samples and time-averaging using a signal-recovery experiment. In both the St. Francis basin and the southeast-Missouri assemblages, drift reproduces the structure without bounded groups. Across the wider valley, it generates far fewer coherent groups than the named phase scheme draws. We find that Parkin, often considered the center of a polity, is a transmission bridge rather than a capital. Tested directly against the other named phases, it shows at most a weak signal at the edge of the drift expectation. A sharper boundary appears only between the central and lower valleys, confounded with chronological turnover and analytical history. Before similarity is interpreted as social, a record must be shown to be able to distinguish groups from drift.

Keywords: cultural transmission, neutral models, frequency seriation, isolation by distance, Mississippian archaeology, Parkin phase

Resumen

La fase, una de las unidades fundamentales de la arqueología, descansa sobre una premisa tácita de que los conjuntos más similares a sus vecinos cercanos que a los distantes registran una sociedad delimitada. Aunque la validez de las fases ha sido cuestionada, la justificación para asumir estas unidades rara vez se pone a prueba. La transmisión cultural neutral produce el mismo agrupamiento sin grupos delimitados, ya que la deriva concentra estilos similares en áreas cercanas. Para el valle central del Misisipi de finales del período precontacto, evaluamos si la similitud de la cerámica decorada registra grupos de interacción delimitados o deriva estructurada espacialmente. Comparamos dos procesos generativos —deriva y grupos delimitados— sobre la geografía real de los conjuntos, calibrando la comparación con tamaños de muestra pequeños y con el promediado temporal del registro mediante un experimento de recuperación de señal. Tanto en la cuenca del St. Francis, como en los conjuntos del sureste de Misuri, la estructura se reproduce mediante deriva y no requiere grupos delimitados; en el conjunto del valle, la deriva genera muchos menos grupos coherentes que los que distingue el esquema de fases mencionado. Encontramos que Parkin, a menudo considerado el centro de una entidad política, es un puente de transmisión más que una capital. Evaluada directamente frente a las demás fases mencionadas, la fase Parkin muestra, a lo sumo, una señal débil en el límite de lo que predice la deriva. Una frontera más marcada aparece solo entre el valle central y el valle inferior, confundida con el cambio cronológico y con la historia analítica. Antes de interpretar la similitud como social, debe demostrarse que un registro puede distinguir los grupos de la deriva.

Palabras clave: transmisión cultural, modelos neutrales, seriación de frecuencias, aislamiento por distancia, arqueología misisipiana, fase Parkin

Introduction

The late pre-contact record of the Lower Mississippi River Valley is traditionally divided into a series of spatially and temporally bounded units known as phases (Figure 1). These units include examples such as the Kent (House 1991), Nodena (Dan F. Morse 1990), Parchman (Starr 1984), Tipton (Smith 1990), Walls (Smith 1990), Jones Bayou (Smith 1990), and Parkin (Phyllis A. Morse 1990) phases. Among these units, the Parkin phase stands out because it includes the Parkin site, the largest of the moated, palisaded towns in the St. Francis basin of northeastern Arkansas. The chronicles of the Hernando de Soto expedition place it as Casqui, the seat of a paramount chief, which the expedition encountered in 1541 (Hudson et al. 1990; Mitchem 2001; Phyllis A. Morse 1990). On the strength of its size, fortification, platform mound, and ethnohistoric identification, Parkin is routinely interpreted as the center of a Mississippian polity, on the long-standing assumption that a dominant mound center marks the apex of a ranked society (Peebles and Kus 1977). That interpretation is a specific claim about social structure in which the Parkin phase is treated as a bounded, hierarchically organized society with Parkin at its apex.

Whether such an entity existed at Parkin is an empirical question. Settlement morphology alone cannot answer this question, since the presence of a primary center of interaction, a palisade, and a monumental mound is equally compatible with competing peer communities as with a centralized, consolidated hierarchy (King and Freer 1995; Rees 2001). The 16th-century chronicles are themselves equivocal, read both as rivalry among competing peers and as a ranked order, so if Parkin is Casqui, they do not, by themselves, establish a consolidated hierarchy over the Parkin phase (Hudson et al. 1990; Phyllis A. Morse 1990). We ask, therefore, what the archaeological record shows when it is measured rather than interpreted through an assumed polity model. Within the basin, settlement size is graded rather than redistributive. The regional production of the capacity-linked chert hoes whose control would mark an economic apex was decentralized, and their exchange followed a down-the-line gradient rather than central redistribution (Cobb 2000). As we show below, the decorated-ceramic record also carries no clear signature of interaction closing in on bounded groups.

The more revealing question is why Parkin was taken for a polity in the first place. It lies in the unit through which the record is interpreted, a matter the renewed debate over archaeological systematics has brought back into focus (Lyman 2021). The Parkin phase is a culture-history construct, a set of assemblages grouped because their decorated-class frequencies seriate together and because the sites cluster in space (Phillips et al. 1951). A social interpretation, however, requires going beyond the phase definition. In the Phillips-Ford-Griffin framework, the phase is the basic space-time-cultural unit, defined by distinctive traits within a locality or region over a brief span, with no social criterion in the definition itself (Phillips and Willey 1953; Willey and Phillips 1958). Willey and Phillips (1958:48–49) even set out the syllogism that would make a phase a society, component equals community, and phase equals a set of components, and then rejected it. In practice, however, though the chances are against archaeological phases having much social reality, they judged them as such and acted as if they did. Williams (1954), defining the southeast-Missouri phases on which the regional scheme rests, adopted the social equation, writing that a phase probably equals a society. That equation, though disavowed at the source, is typically repeated in regional practice. This assumption about the unit phase is its silent premise. It holds that assemblages more similar to one another than to their neighbors record a bounded, mutually interacting social group. In the case where one can identify a group with a primate center, the phase then becomes a polity. The polity interpretation is built into the unit before any independent evidence of bounded groups is weighed.

The phase construct has been questioned on exactly this ground. Fox (1998) showed, for example, by cluster analysis and ordination, that the southeast-Missouri phases fail as both groups and classes, with their member assemblages assigned to statistical clusters at random rather than segregating by phase. Mainfort (2003) reached the same verdict by a different ordination, finding that the ordination of the late-period assemblages provides little support for the validity of the named phases, which overlap rather than forming discrete clusters. Lipo (2001) questions the nature of phases through frequency seriation, showing that the assemblages are structured rather than panmictic, and arguing that the resulting spatial groups have no warrant as societies. He (2001:60) suggests "the phases themselves might be simply arbitrary spatial and temporal slices of a single interacting population that is much larger in size."

The silent premise, however, is tested only rarely and overlooks a strong alternative that offers a potentially falsifiable explanation for the observed patterns in the archaeological record. Viewed through the lens of social learning and models of cultural transmission, interaction is always continuous yet highly variable. Interaction is structured by distance and geography, as well as by social relationships. The task is to show how apparent differences between groups, such as phases, can arise under continuous interaction rather than requiring bounded groups. Under the neutral model of cultural transmission, decorative style behaves as a selectively neutral trait whose frequency drifts in a finite population of interacting learners (Bentley et al. 2004; Dunnell

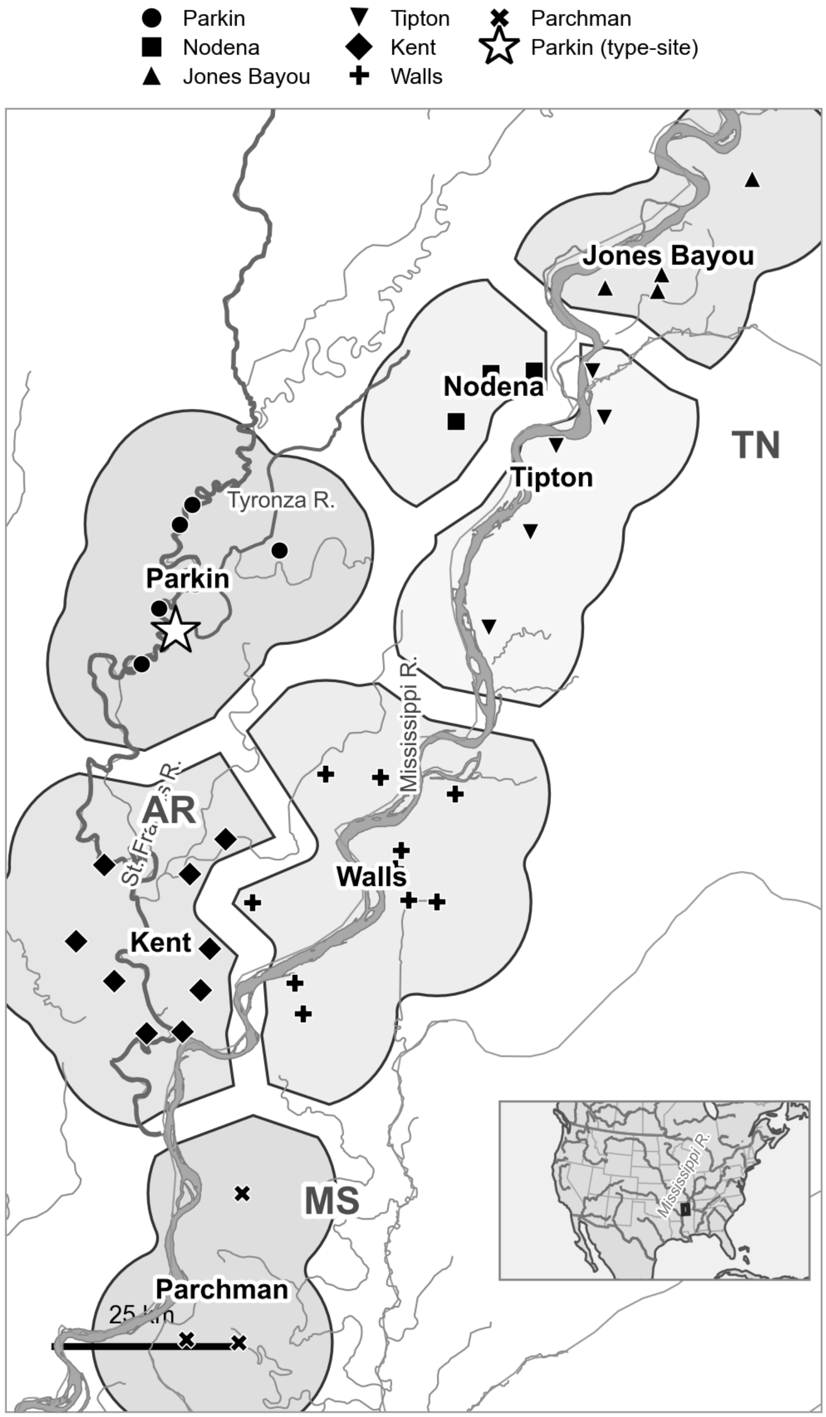

Figure 1. Decorated-ceramic assemblages of the central Mississippi Valley and the culture-historical phase scheme this paper tests. The seven Late Mississippian phases (Parkin, Nodena, Jones Bayou, Tipton, Kent, Walls, and Parchman) are shown as discrete territories, a Voronoi partition of the assemblages Mainfort (1996b) assigns to each phase, dissolved by phase and drawn with a gap between adjacent territories for legibility, over a local hydrology base (the St. Francis, Tyronza, and Mississippi rivers) with state and county boundaries. Each assemblage is plotted with a marker corresponding to its phase, and Parkin is marked with a star. The bounded territories illustrate the premise that the paper tests, that such areas host distinct, internally interacting societies. The transmission analysis finds instead that drift structured by geography explains the impression of groupness. The mapped assemblages fall in Arkansas, Tennessee, and Mississippi; Missouri lies just north of the frame. Inset: location in North America.

1978; Eerkens and Lipo 2007; Fogarty et al. 2024; Neiman 1995). When those learners interact more with near neighbors than with distant ones, drift on the spatially structured population generates similarity that is highest at short distances and fades with distance. Viewed at larger scales and divided into discrete units, that gradient looks like difference and clustering, with no bounded groups anywhere (Lipo et al. 1997, 2021). Based on this account, a phase can be the ordinary residue of continuous, distance-structured interaction. The very pattern the culture-history tradition interprets as a society, assemblages near each other in space and in style, is what spatially structured drift predicts. Telling a real interaction group apart from spatially structured drift requires a test that the tradition never applied, and that is not possible with culture history methods, because the seriation coherence and spatial proximity that define the phase are equally the products of drift (Riede et al. 2019).

Decorated ceramics provide an instrument for the test. Because stylistic variants are copied among potters through social learning networks, the spatial and temporal structure of decorated-class frequencies reflects past interactions. A genuine social boundary, had one existed, would make learning appear conformist within groups and divergent between them, fragmenting the single drifting pool that continuous interaction maintains (Lipo et al. 1997, 2010, 2015; Neiman 1995). The late pre-contact central Mississippi Valley is a favorable place to look, both because its decorated assemblages are well documented and because European contact and the subsequent demographic collapse ended the sequence before any later reorganization could overwrite it (Morse and Morse 1996). We first focus on the St. Francis basin, the setting of Parkin (Figure 1), and then extend the test to other late-period phases of the valley.

The argument rests on a choice about what the ceramic record is. We do not treat an assemblage as a degraded snapshot of a past behavioral moment, to be restored by correcting for the processes that formed it (Schiffer 1972). We treat it as what it is: a time-averaged accumulation whose decorated-class frequencies are the population-level residue of social learning, to be measured and explained directly rather than reconstructed (Lipo et al. 2015; Neiman 1995). This is an evolutionary position in which the record is the evolving phenotype to be explained rather than a window onto a vanished behavioral system (Dunnell 1978, 1980, 1995a; Ladefoged and Graves 2000), and it converges with the recognition that the deposit is not a frozen tableau (Binford 1981). The record's coarse temporal resolution is also not a defect to be removed, but the scale at which long-term, population-level processes become legible (Bailey 2007; Behrensmeyer et al. 2000; Holdaway and Wandsnider 2008; Perreault 2019; Stern 1994). Time-averaging sets the resolution of the question rather than disqualifying it. A reorganization of transmission large enough to matter over the two centuries of the Parkin phase should register in the averaged frequency structure. Thus, a drift-versus-groups test must measure that structure directly rather than be based on an imagined moment that is assumed to be behind it.

No single transmission signature can settle the question, because each is consistent with several processes. A departure from neutral copying, a spatial cline, or a rise in between-group variance can arise from drift, conformity, or environmental patchiness alike. This is an example of equifinality, in which the same pattern can be reached by different processes, undermining the utility of one-signature inferences (Barrett 2019; Crema et al. 2016; Kandler and Shennan 2015; Mesoudi 2021). To avoid this problem, we compare multiple processes in a generative manner by simulating neutral spatial drift and bounded groups on the real coordinate layout of the assemblages. Using simulated data, we ask which measures can reproduce the observed structure, using four transmission signatures: departure from neutral copying, breakdown of frequency-seriation coherence, between-group variance, and spatial assortativity. Because archaeological assemblages are time-averaged and sampled, our measurements may potentially reflect differences that are artifacts of sampling rather than of transmission. Consequently, we calibrate our comparisons to the record using a signal-recovery experiment in which synthetic assemblages are matched to the real number, sample sizes, and time-averaging. We then inject a known signal of tunable strength and measure how strong a real boundary would have to be to register. This calibration enables us to conclude that the presence or absence of group structure is real, not a failure of resolution. This calibration is the transferable contribution, because before any similarity structure is interpreted as social, a record must be shown to be able to distinguish drift from groups at all.

The argument proceeds in three steps. First, we evaluate the Parkin as the topmost part of a hierarchy and find that no line of evidence, settlement, exchange, or transmission supports such a claim. Second, we ask what generates the St. Francis basin phase structure, and show that neutral drift on the basin's geography reproduces the observed signatures. A bounded-groups process would leave a far sharper structure than the data demonstrate. Hence, we conclude that the phase is a portion of a continuous interaction structure rather than a closed group with a boundary, and Parkin is a transmission bridge rather than the core of a group. Third, we widen the test to the other late-period phases of the valley and ask whether decorated-ceramic boundaries exist. A boundary that exceeds neutral drift appears only between the central-valley and lower-valley blocks, and even there, it is confounded with the early-to-late turnover of the class repertoire. It cannot yet be interpreted as a synchronic social division. At the scale where the phase is drawn, it is largely indistinguishable from spatially structured drift. We ultimately conclude that Parkin is best understood as a prominent node in a continuous field of interaction rather than the capital of a polity.

Background

The phase, the silent premise, and the drift alternative

In evaluating the phase concept, we take a materialist approach to cultural transmission, one that frames the question differently from how the Lower and Central Mississippi Valley records are usually read. Archaeologists often treat the past as made of bounded units, the phase, the polity, the society, each a discrete entity with members and edges, and read a cluster of similar assemblages as one such unit. We treat transmission instead as a continuous, variable field of copying among potters, with no intrinsic boundaries. Interaction falls off with distance and is more local in some places and times than in others, so a group is not a thing with a boundary but a region where copying is locally intensified, and similarity is a matter of degree. Closure is a matter of degree, too: interaction can narrow until potters copy mostly within a subset and rarely outside it, producing something that resembles the bounded groups archaeologists assume.

Operationally, a phase is a set of assemblages that seriate together and cluster in space. It is interpreted, however, as a social entity, a bounded interacting community, on the premise that assemblages more alike within a phase than across its edges record individuals who interacted mostly within (McNutt 1996; Morse and Morse 1983; Phillips et al. 1951; Williams 1954). That premise is not unreasonable. The difficulty with accepting this interpretation, however, is that the same clustering arises with no boundary at all, through isolation by distance (Lipo et al. 2021; Neiman 1995; Shennan et al. 2015). What would mark a real bounded group is closure beyond what distance predicts, and that excess is the signal we must test for. We can treat decoration transmitted preferentially within a group and divergent between groups as a marker, rather than a neutral style drifting in a shared pool. A marker is the ceramic proxy for interaction closing into a bounded group, so the drift-versus-groups question is effectively a question of neutral style versus marker.

Why ceramics, and the neutral-vs-marker distinction

Measures of variability in decorated ceramics can capture the structure of social interaction because stylistic variants are copied among potters through social learning networks. Under the neutral model of cultural transmission, stylistic variants behave like selectively neutral alleles whose frequencies are governed by innovation and drift in a finite population of interacting learners (Bentley et al. 2004; Bentley 2008; Dunnell 1978; Eerkens and Lipo 2005; Neiman 1995). Assemblages drawn from a single interacting community share one drifting pool, and their decorated-class frequencies form a single battleship-curve seriation (Dunnell 1995b; Lipo 2001; Lipo et al. 2015; Lipo and Madsen 2001; O'Brien and Leonard 2001). Assortment, the tendency for like to learn from results in interaction, sorts into distinct groups and alters patterns of frequency changes. If learning becomes restricted within emerging social boundaries, decoration shifts from the expectations of neutral, stylistic variation toward a constrained set of possibilities, transmitted by conformist copying within groups and divergent between them, and the single transmission field fragments (Crema et al. 2016; Henrich 2001; Henrich and Boyd 1998). When this process plays out, the restricted pool of variation can come to look like "markers" of bounded social groups, even if no sharp bounds existed.

A methodological obstacle, however, is that no single frequency pattern is uniquely associated with its generating process. A departure from neutrality can reflect conformity, an anti-conformist novelty bias, a content bias, or simply the fact that durable objects, rather than people, are the units that accumulate in a deposit (Crema et al. 2016, 2024; Henrich 2001). The problem is more complex in archaeological assemblages, which are time-averaged because deposits form through aggregation over a use-life, a process that inflates richness, flattens evenness, and biases neutrality statistics by an amount that depends on the mean lifetime of the variants (Madsen 2012; Premo 2014). The result is equifinality, in which multiple transmission processes produce the same population-level pattern. Thus, a single statistic cannot recover the process that generated it. A discriminating signal, where it survives, only lies in the rare-variant tail of the frequency distribution rather than in its common classes (Carrignon et al. 2019; Kandler et al. 2017; Mesoudi 2021; O'Dwyer and Kandler 2017).

The central Mississippi Valley case and two competing hypotheses

The late pre-contact central Mississippi Valley supports two interpretations of the same record, and they make opposite predictions for the transmission signature. The first holds that the region was moving toward greater complexity. In northeastern Arkansas, the Parkin phase (ca. 1400 to 1600 CE, with a thin protohistoric tail consistent with the basin radiocarbon reported below) is a dense cluster of fortified, moated St. Francis-type towns along the St. Francis and Tyronza rivers. It is dominated by the Parkin site, a roughly 17-acre palisaded center whose platform mound rises more than 6 m, which the De Soto chronicles place as the seat of the paramount of Casqui (Dye and Cox 1990; Mitchem 2001; Phyllis A. Morse 1990). Based on this information, Parkin's location on agriculturally poor soil has been interpreted as evidence that it drew tribute from smaller villages, a hallmark of an emerging chiefly hierarchy (Phyllis A. Morse 1990). European contact would have truncated any such transition before it could consolidate, the florescence Morse and Morse describe as cut short "by the appearance of De Soto and his party" (Morse and Morse 1996:17–18). We treat this truncation as a working premise rather than a settled fact, because the basin's late sequence is poorly dated (Morse and Morse 1996). The second interpretation holds that the central valley was a stable arrangement of competing peer polities that never consolidated into a regional hierarchy (Anderson 1996; Beck 2003; Blitz 1999; King and Freer 1995; Lulewicz 2019; Milner 1998; Rees 2001). In the chronicles, Casqui and its neighbor, Pacaha, appear in open conflict, consistent with those of competing peers. The same sources have also been interpreted as ranking Casqui below Pacaha, and the chronicle ethnography is thin and filtered (Hudson et al. 1990; Mitchem 2001; Phyllis A. Morse 1990). The two possibilities for Parkin cannot be separated by settlement morphology alone. They can, however, be separated by determining whether the transmission record shows interaction closing into bounded groups, which is the test we now construct.

Distinguishing true group structure from spatially structured drift

Our test for detecting group structure rests on a single principle: a process that genuinely reorganizes a population into groups must leave a mark on every aspect of how pottery styles were copied, whereas a process that merely mimics group structure leaves a mark only on some aspects. The reorganization at issue, which we call the closure of interaction, is central to the rest of the paper, so we define it intensionally (*sensu* Dunnell 1971). In an open community, potters can learn a design from anyone they encounter, near or far, so styles circulate within a single connected population. Interaction closes when that circulation narrows. Potters mostly copy others within their own group and rarely anyone outside it, until a once-continuous field of social learning breaks into separate pools that drift apart. Closure is the process a genuine bounded interaction group undergoes, and we treat its strength as a quantity that runs from none (one fully open field of learning), through partial, to complete (sealed, non-interacting groups). That strength is what we set out to test and to measure. In the synthetic experiments below, we treat it as a parameter, s, from 0 (pure drift, no closure) to 1 (strongly bounded, sharply separated groups). We can evaluate four potential signatures, each a different consequence of closure, and we state each in plain terms before naming their formal measures.

The first asks whether potters copied decorated styles at random or with a bias. Under unbiased copying, where a potter is as likely to reproduce any design she encounters as any other design, there are two independent ways to assess agreement in assemblage diversity. One is based on homozygosity, a measure borrowed from genetics. This property reflects the probability that two sherds drawn at random from an assemblage share the same decorated class, which is high when a few classes dominate. The other is based on richness, the number of classes in which sherds are identified in an assemblage. Each yields an estimate of the same underlying transmission parameter, and under random copying, the two estimates coincide. Conformity, in which potters disproportionately copy whatever is already common, concentrates frequency onto a few classes. This raises homozygosity but barely changes the number of classes present, because rare designs persist at low frequency. The two estimates then diverge because they respond to different things. The homozygosity-based estimate runs opposite to homozygosity and thus declines, while the richness-based estimate tracks the nearly unchanged class count and holds steady. A taste for novelty does the reverse. The disagreement between the two estimates, expressed as their ratio, therefore detects a departure from random copying and indicates the direction of that departure (Neiman 1995; Shennan and Wilkinson 2001). This matters here because a bounded group that enforced a shared decorative marker would copy conformist behavior even to the edges of its boundaries. As a result, the two estimates would diverge, whereas a single pool of styles drifting at random would keep them equal.

The second asks whether the assemblages all belong to one interacting community or to several, and it is the most direct test of fragmentation. It rests on frequency seriation, the classic archaeological method of ordering assemblages so that each decorated class's relative frequency rises to a single peak and then falls away, the shape long known as a battleship curve (Dunnell 1970). That single-peaked rise and fall is how stylistic traits behave: they are introduced, become popular, and are abandoned within a community that draws on a shared pool of designs. In classical uses of seriation, all available assemblages were typically pooled into a single seriation, and deviations from the frequency guidelines were dismissed as sampling error (Dunnell 1970). Sampling error, however, offers valuable information, though classical practice tends to ignore this most valuable aspect of the method (Lipo 2001). A *failure to seriate together* indicates that locales are far enough "out of step" with the surrounding flow of interaction and social learning that their interaction history has interesting structure (Lipo et al. 1997, 2015).

A set of assemblages is "co-seriable" when a single ordering can make every class follow the same single-peaked shape simultaneously. When all assemblages arise from a single continuously interacting population, such an ordering exists. When learning splits into separate groups, each with its own repertoire and its own independent rise and fall of styles, no single ordering can satisfy every class at once, and the assemblages break into several mutually incompatible orderings. The deterministic frequency-seriation procedure of Lipo, Madsen, and Dunnell (2015) finds the complete set of these maximal co-seriable groups algorithmically, without the analyst arranging the assemblages by eye (Lipo et al. 1997). Using this approach, the number of groups and the number of groups any one assemblage joins directly measure the structure of past interaction. A single large group means one connected field of learning, a proliferation of small groups means a fragmented one, and an assemblage that falls into several incompatible orderings is a bridge linking otherwise separate sequences rather than the closed core of a bounded community.

The third asks how much of the variation in decoration lies between groups rather than within them. The measure is a cultural version of Wright's F_ST (Wright 1951), a standard population-genetic index of group differentiation. In cultural transmission, between-group variation in socially learned traits routinely exceeds that of their genetic counterparts (Bell et al. 2009). We measure decorated diversity with the Gini-Simpson index, the probability that two sherds drawn at random from an assemblage are of different classes, computed both within each spatial group and across all pooled assemblages. F_ST is how much larger the pooled diversity is than the average within-group diversity, expressed as a fraction of the pooled diversity. When every assemblage draws from one shared, drifting pool of styles, pooling adds little diversity beyond what is already present within each group, and F_ST sits near zero. As groups come to favor distinct repertoires, the between-group share grows, and F_ST climbs (Neiman 1995). F_ST serves here as the most direct measure of how far decoration has sorted itself by group, with one important caveat developed below. Stochastic drift on a spatially structured population generates between-group variance on its own. Subpopulations retain different levels of diversity depending on their size and connectivity. Small or sparsely connected areas drift toward depleted, idiosyncratic repertoires, while larger, well-connected ones remain richer. The resulting differences appear as distinct interacting groups, even though they are produced solely by spatial position (Lipo et al. 2021). The measure must therefore be calibrated against that spatial-drift null rather than taken at face value.

The fourth asks whether socially close assemblages are more alike than assemblages that are merely nearby on the map. Similarity normally fades with distance, because potters interact more with near neighbors than with distant ones, a gradient known as isolation by distance (Neiman 1995; Shennan et al. 2015; Wright 1943). A genuine social boundary adds a step to that gradient. Under this condition, pairs of assemblages within the same cluster are more similar than pairs in neighboring clusters, even when the two pairs span the same geographic distance. We capture that step as a boundary-excess statistic, which measures the gap between within-cluster and between-cluster similarity once distance is held constant, a metric that a plain correlation of similarity with distance cannot isolate. It is useful here because this measure separates a real social edge from the ordinary spatial fade, again with the caveat that drift on a finite network can itself produce a positive excess (Lipo et al. 2021).

Each plausible alternative to a bounded interaction group involves only a subset of these signatures (Table 1). Environmental patchiness, in which similar people occupy distinct ecological zones, produces spatial structure with no departure from neutral transmission and no loss of seriation coherence. Aggregated signaling, the bottom-up regime in which a conformist bias flattens diversity within a single community, produces a departure from neutrality without boundary formation, so seriation remains coherent and between-group variance remains low. Smooth, deterministic drift with distance-limited interaction produces an isolation-by-distance cline (Shennan et al. 2015; Wright 1943). This pattern appears as a gradual fading of similarity with distance, as people interact mostly with near neighbors, without the sharp boundaries or departures from neutrality that mark restricted, assortative learning. A bounded interaction group, in which one assortative process restricts transmission along every dimension, is the only process that would drive all four signatures together at full strength.

Table 1 presents an idealized comparison of these signatures. Two of the four signatures, between-group variance and the distance-controlled boundary excess, are also driven by stochastic neutral drift on a spatially structured population, which generates patchy divergence and a positive boundary excess with no bounded group anywhere (Results, Figure S3). The linked movement of these two measures cannot separate a real interaction group from drift. Thus, we cannot depend on these signatures alone. Instead, we must compare the two processes generatively by simulating each one, neutral spatial drift and bounded groups alike, using the real geography of the deposits. The simulations provide a means of determining which one reproduces the observed structure, treating the signatures as diagnostic of which process left its mark.

Table 1. What each process does to the four transmission signatures, and which signature this record can measure. A filled circle (•) marks a signature, while an open circle (◦) is one where it is not measurable. The process columns give the idealized discrimination logic. Under smooth, deterministic processes, only a bounded interaction group drives all four, and each alternative drives a characteristic subset, the pattern an idealized simulation confirms (Figure S1). Two features of the table show why that logic is not the test we run on the record. The smooth-drift and stochastic-drift columns separate textbook isolation-by-distance, which drives none of the four, from realistic neutral drift on a finite network, which drives cultural F_ST and the boundary excess on its own, with no bounded group present (Figure S3). Stochastic drift thus reproduces the same two-signature pattern as environmental patchiness, so a positive result on either signature cannot by itself separate a bounded group from drift or patchiness. The final column reports the recovery experiment introduced below. The four signatures are candidate diagnostics screened against the record, and the surviving discrimination is carried by cultural F_ST, assessed against the drift null, and by the generative drift-versus-groups comparison, not by convergence of the table's rows.

Signature	Bounded group	Environmental patchiness	Aggregated signaling	Drift (smooth)	Drift (stochastic)	Detectable on this record?
Departure from neutral transmission	•	◦	•	◦	◦	No
Seriation-coherence breakdown	•	◦	◦	◦	◦	No

Signature	Bounded group	Environmental patchiness	Aggregated signaling	Drift (smooth)	Drift (stochastic)	Detectable on this record?
Between-vs within-group variance (cultural F_ST)	●	●	○	○	●	Yes
Spatial assortativity (boundary excess)	●	●	○	○	●	No

The two drift columns illustrate the difficulty archaeologists face in detecting bounded groups. Smooth, deterministic isolation-by-distance produces only a pattern of decay in similarity with distance. In addition, the distance-controlled boundary excess is near zero under it, so it does not drive any of the four signatures. Stochastic neutral drift on a finite network is less forgiving. It generates patchy divergence that raises cultural F_ST and registers as a positive boundary signal even in the absence of a social boundary, a mimicry that the record-matched controls demonstrate (see Results, Figure S3). Aggregated signaling, by contrast, raises the neutrality departure within a single undivided pool without forming the between-group boundaries that the other signatures require.

If closure were to occur, it would accumulate across the sequence. On idealized data, genuine convergence should take the form of a trajectory, a coherent, same-direction shift in all four signatures across the ordinal sequence of assemblages, concentrated where and when interaction closes into bounded groups. The logic, however, is only as strong as the independence of the signatures, because four measures that were merely redundant converge trivially. As a consequence, we validate the test using simulated assemblages before applying it. This validation seeks to confirm that convergence occurs only when interactions cluster into a bounded group, and not under any other condition. We also use the simulation to assess whether the four signatures are tightly correlated under such closure, yet nearly independent under alternative scenarios. In this way, we can be assured their joint movement will be informative. The use of convergence logic has an archaeological precedent. Neiman (1995), for example, showed that two independently derived, theory-grounded measures of the same ceramic data, within-assemblage diversity and between-assemblage distance, converge on the same inference about past interaction. We generalize his approach and apply it to our four measures, ensuring dynamic sufficiency. Empirical sufficiency is another question. Whether all four signatures, or only some, can actually do so on a record like ours is a separate question that we settle by using a calibration test that evaluates synthetic data matched to the Parkin assemblages.

The calibration process consists of a signal-recovery experiment. We inject a known group-closure signal of controlled strength into synthetic assemblages matched to the real record, then ask which signatures recover it and how strong the closure must be to register. Reading each signature against this record-matched standard, rather than against the idealized expectation, is what enables us to treat an absence of structure as a real result rather than a failure of resolution. The procedure and its parameters are detailed in the Materials and Methods section and the Supplemental Text.

Below, we describe our procedures, methods, and outcomes. Figure 2 previews what the test distinguishes. It shows the sharp structure that a bounded group would leave against the diffuse pattern that drift produces, and why, on a record of this size and time-averaging, a weak within-basin boundary cannot be excluded.

A signature is trustworthy to the degree that it meets three conditions, usually treated apart. It should be solidly based in theory. In this case, a cultural transmission model provides its assumptions, resulting in a

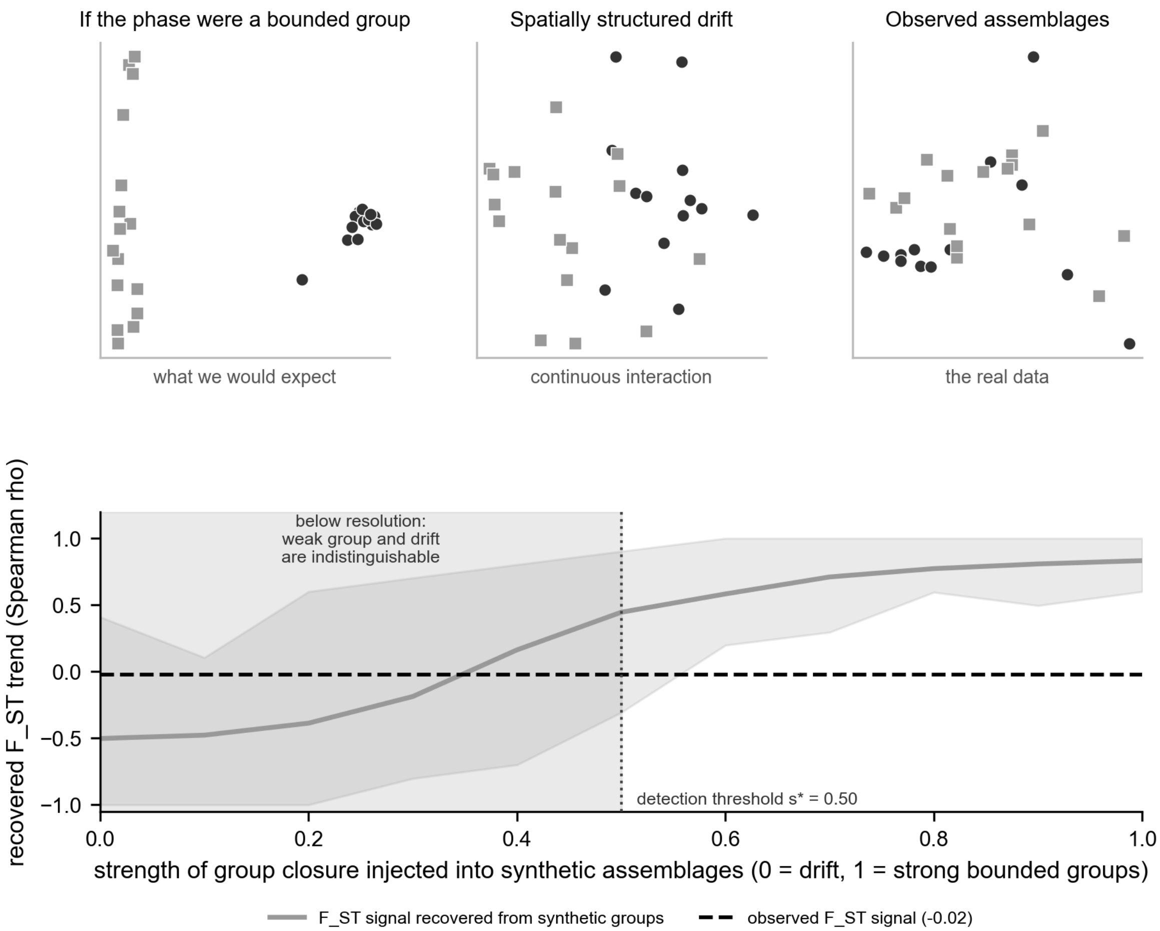

Figure 2. What the test distinguishes, and why the within-basin determination is hard, from synthetic assemblages matched to the record. Top: the basin assemblages in a multidimensional scaling of decorated dissimilarity (points colored by the k=2 spatial split) as they would appear if the Parkin phase were a bounded interaction group (left, sharp clusters), under spatially structured neutral drift (center, a continuous gradient), and as observed (right, a diffuse scatter resembling drift). Bottom: the cultural F_ST signal recovered from synthetic assemblages as the strength of injected group closure rises from 0 (drift) to 1 (strong groups), with the s=0 null band shaded behind it. The signal clears the detection threshold only at $s^* = 0.5$. The observed value (dashed line) lies in the shaded region below s*, where a weak group and pure drift are indistinguishable at this record's size and time-averaging. This is why a weak within-basin boundary cannot be excluded.

theoretical claim rather than an analogy. Deterministic frequency seriation is the model case, because the unimodality and continuity it requires are exactly what the neutral model predicts of a single interacting community. Neiman's (1995) diversity and distance estimators are likewise derived from the same cultural transmission model (Lipo et al. 2015). It should be dynamically sufficient in Lewontin's (1974) sense, with its state variables and their transformation laws sufficient to carry the system from one state to the next. And it should be empirically sufficient. This condition means it is matched to the quality of the archaeological record and defined at the time-averaged resolution preserved by the deposit, rather than at a snapshot or generational resolution (Holdaway and Wandsnider 2008; Perreault 2019).

Thus, our four signatures are theoretically and dynamically served by the transmission model. Empirical sufficiency, whether a signature can actually register a signal at the resolution the deposit preserves, is not guaranteed by this framing and requires us to test it rather than assume it. As we will describe below, at this record's 29 time-averaged assemblages, only cultural F_ST registers the injected signal, and even then, it must be compared against the stochastic-drift null rather than zero. Seriation-coherence breakdown, which only a bounded group drives in principle, is the cleanest discriminator on paper yet undetectable here. Measured as states along the sequence, the signatures are only partially dynamically sufficient, a gap we close in the discussion with tests that model the transformation between states.

Materials and methods

Data

The ceramic data that are the basis of the original phase definitions come from the decorated-class counts compiled by Phillips, Ford, and Griffin (1951) for the Lower Mississippi Valley Survey. We have compiled these data into an assemblage-by-class matrix of 266 surface assemblages (Lipo 2001). Each decorated class is a culture-history type, a unit of measurement that the analysts defined to capture stylistic variation, rather than a category assumed to exist in the past. For our study, we restrict the analysis to the St. Francis basin of northeastern Arkansas, the setting of Parkin, defined hydrologically as the assemblages within 20 km of the St. Francis, Tyronza, and L'Anguille drainage (Figure 1). This yields 29 decorated assemblages for the transmission analysis. The basin is a geographic unit, not a single culture-historical phase. Under the standard regional scheme, its assemblages fall into the Parkin phase together with sites assigned to the Nodena, Kent, Walls, and Parchman phases (Mainfort 2003; Phillips 1970). The test asks whether interaction closed into bounded groups anywhere in the basin, across the phase divisions that culture history draws within it, and within the Parkin phase. The set excludes the lower Yazoo Basin to the south. A stricter watershed rule, keeping only assemblages nearer the St. Francis system than the Mississippi, gives 19. The verdict reported below is unchanged across both definitions. The two southern St. Francis-type sites the drainage rule excludes, Old Town and Blanchard, lie 28 and 80 km from the St. Francis system and belong to the Mississippi-proximal and lower Yazoo settings rather than to the St. Francis drainage. We georeferenced each assemblage based on maps from the Lower Mississippi Survey site files. We used settlement attributes, site size, mound count and height, and the presence of defensive ditches and St-Francis-type fortification, from a regional geographic database (Lipo and Dunnell 2007).

Two properties of the data bound what any transmission analysis can claim, and we treat them as constraints rather than nuisances. First, the assemblages are time-averaged surface assemblages, so they consist of a sample of an aggregate of pottery formed over the use-life of its deposit. Such aggregation distorts diversity and neutrality statistics in proportion to the mean lifetime of the variants (Madsen 2012; Miller-Atkins and Premo 2018). Second, the neutral model's expectations are derived for a population of people copying one another. An assemblage, however, counts durable objects, the pots and sherds themselves, not the potters who made them. Because the record is object-mediated rather than person-mediated, the standard neutral expectation does not apply exactly and must be understood with that caveat in mind (Crema et al. 2024).

The availability of an absolute chronology is also a requirement. A compilation by Mainfort (2001) provides 109 determinations for this part of the Mississippi valley, 41 of them from seven St. Francis basin Parkin-phase sites and 19 from Parkin itself (House 1993; Mainfort 1996a; Sullivan et al. 2024). Calibrated against IntCal20 (Reimer et al. 2020), the basin determinations place the occupation in the fourteenth to sixteenth centuries (summed-probability median AD 1422, Parkin AD 1483) and show a decline across the contact interval, as detailed in the Supplemental Text. A calibration plateau across that interval defeats year-scale resolution (Holland-Lulewicz and Ritchison 2021), so we order the assemblages by seriation rather than by calendar age, using the first axis of a correspondence analysis (Baxter 1994; Peeples and Schachner 2012; Smith and Neiman 2007). This ordination arranges assemblages along axes that summarize their decorated-class-frequency differences, as a relative chronological ordinate oriented so that higher positions correspond to later time (Figure 3). Pooling the dates of the five proveniences that match curated assemblages, the median calibrated ages correlate with seriation position at Spearman +0.70 (n = 5, 27 dates pooled). Four of the five sites fall in calendrical order, with one late-dated site departing from it. The axis tracks calendar time imperfectly. What we test is an ordinal trajectory, not a calendrical rate. The discriminating quantity, the size-controlled F_ST trend, is a magnitude invariant to the axis orientation, so only the directional descriptions of the other signatures depend on the anchor. We return to both limits in the discussion.

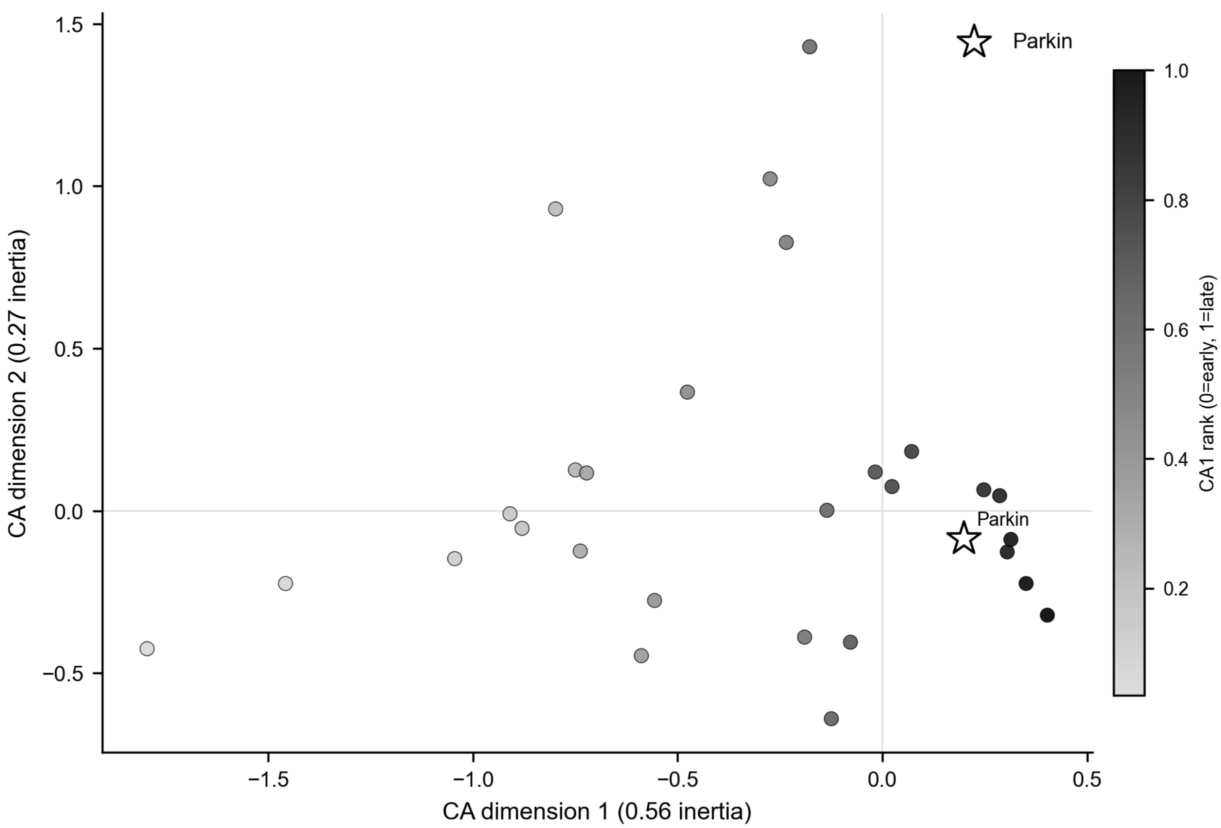

Figure 3. Correspondence-analysis seriation of the 29 basin decorated assemblages, plotted on the first two CA dimensions of the decorated-class frequencies and colored by CA1 rank (the relative chronological ordinate, oriented by the five available radiocarbon dates so that higher rank is later). Parkin is starred. The assemblages form a gradational continuum rather than discrete clusters, echoing the independent NMDS result of Mainfort (2003).

The four signatures

We computed each signature with a published or standard estimator and combined them only at the final step. The first signature, departure from neutral transmission, follows Neiman's (1995) application of the infinite-alleles model to ceramic assemblages. The transmission parameter is estimated from an assemblage's unbiased homozygosity and, via the Ewens sampling formula (1972), from its number of classes, and the ratio of the two is the signature (Shennan and Wilkinson 2001). Because pooling assemblages from divergent groups cancels the signal, we computed it within spatial clusters and averaged across them.

The second signature, seriation coherence, follows the iterative deterministic frequency-seriation method of

Lipo, Madsen, and Dunnell (2015), which returns the complete set of maximal co-seriable groups. Because checking every possible ordering becomes computationally intractable beyond about fourteen assemblages, IDSS builds solutions agglomeratively, starting from the valid three-assemblage seriations and extending each by adding assemblages that keep every class unimodal and continuous, until none can be added. Each class's departure from a single peak, or a break in its continuity, is assessed against bootstrap confidence intervals for the class frequencies, so it counts as a real break only when it exceeds the sampling error. Small, sparsely sampled assemblages have wider intervals and admit more orderings. We report the number and size of these groups and, for each assemblage, the number of distinct groups it joins.

The third signature, the variance partition (cultural F_ST), is computed from Gini-Simpson diversity across the spatial clusters, with within-group diversity size-weighted by assemblage count; the exact estimator is provided in the Supplemental Text (Bell et al. 2009).

The fourth signature, spatial assortativity, is computed from the Brainerd-Robinson similarity of each assemblage pair (Brainerd 1951; Robinson 1951), a standard index from 0 to 200 on class proportions. We test its association with inter-site distance with a Mantel (1967) permutation test and, because a Mantel correlation cannot separate a sharp boundary from smooth distance decay, also compute the distance-controlled boundary excess.

Two corrections are essential before any of these signatures is interpreted. Assemblage sample size trends with seriation position (Spearman's rho = +0.59), and several estimators are size-sensitive, so we subsample every assemblage to a common count (rarefaction) (Hurlbert 1971) before scoring each signature as its rank correlation with ordinal position. Which signatures can then be trusted at the record's resolution is not assumed but established by the recovery experiment described next, in which a known closure signal is injected into the real configuration.

Criterion validation by simulation

We ran four simulations, each answering a different question. The first is an idealized validation. It asks whether the four signatures can, in principle, separate a bounded interaction group from its three mimics, using large, clean synthetic assemblages (Figure S1). The second is a record-matched recovery. It asks which of those signatures can still detect a known, injected closure once the synthetic assemblages are held to the real record's sample sizes and time-averaging (Figure 4). The third is a positive control. It asks which generative process, neutral spatial drift or bounded groups, reproduces the structure actually observed in the basin (Figure S3). The fourth is a forward demonstration. It runs neutral drift on the real site geography with no boundary imposed, counts how many phase-like groups it produces, and checks whether the named phases exceed that number (Figures 8 and 9). The first two simulations establish what the test can resolve on a record of this size. The second two simulations identify which process the data match. We describe the validation and recovery here, and the two drift simulations in the next subsection.

We validated the test in two stages, an idealized validation and a record-matched recovery. The idealized validation rests on a generative model of decorated-variant transmission under neutral copying, in which learners reproduce variants in proportion to their current frequency (Bentley et al. 2004; Neiman 1995). The model has three tunable controls: the between-group divergence of class profiles, the strength of within-group conformity, and a spatial rule placing group profiles on the landscape. Four generators instantiate the four idealized processes of Table 1, the bounded group and its three mimics (Mesoudi 2021). We ran the simulation blind through the four-signature procedure across twenty random seeds. Only a bounded interaction group drives all four signatures together, and the signatures are tightly correlated under closure yet nearly independent under the mimics, so their joint movement is informative rather than an artifact of redundancy (Supplemental Text, Figure S1). The idealized validation uses large, clean assemblages and shows what the test can do in principle, not what it can do on a record like ours.

The record-matched recovery builds synthetic pottery records in which we control whether group structure is emerging, dialed from none to strong, and ask whether our four measures can see it. The synthetic records were matched to the real ones. We create sets consisting of the same number of assemblages, the same range of sample sizes, and the same time-averaging imposed by surface assemblages (Madsen 2012; Miller-Atkins and Premo 2018; Premo 2014). If chiefdom structure had really been emerging at Parkin, this experiment would tell us whether methods like ours, applied to data like ours, would have caught it. Concretely, we generated synthetic assemblages, one per real assemblage, at the real per-assemblage sample sizes, spatial clusters, and ordinal bins. Into them, we injected a known group-closure signal of tunable strength s, with between-group divergence and within-group conformity rising together along the sequence to a maximum of s. We then asked which signatures recover it, a record-matched recovery design in the spirit of simulation-based inference calibrated to the data at hand (Crema et al. 2014; Kandler and Shennan 2015). Two properties of the record make this necessary. The size-seriation trend noted above means a size-dependent estimator can move along the axis through sampling alone, so we subsample every assemblage, synthetic and real, to a common count before scoring. It is also time-averaged, which we reproduce by blending adjacent-bin production within a window. The recovery experiment then asks, at the resolution and sample sizes this record preserves, which signatures can be trusted and how strong a closure would have to be to register. One aspect of the calibration matters here. Because subsampling the larger, later assemblages to a common count slightly inflates within-cluster diversity and so depresses F_ST toward later positions, the null with no closure (pure drift) itself carries a mild negative trend rather than a flat zero. The test is calibrated against that null distribution, not against zero, so the false-positive rate is held at five percent regardless, and the empirical value is compared against the same null.

The drift-versus-groups comparison

The central test compares two generative processes on the real geography of the assemblages. The neutral spatial-drift model follows the finite-population network model of Lipo, DiNapoli, Madsen, and Hunt (2021). Each assemblage is a node holding a finite population of learners that copies decorated-class variants from within the node and, at a low rate, from other nodes with a probability that decays with geographic distance. Innovation enters at a fixed rate, and deposition is pooled over a trailing window to reproduce the record's time-averaging. The bounded-groups model is identical except that interaction is concentrated within a small number of spatially contiguous groups, with only a low leak of copying between them, so that learning is restricted within boundaries. Both are run on the observed coordinate layout and subsampled to the observed per-assemblage sherd totals. Both are scored by the same four signatures, along with network modularity, a measure of how strongly the assemblages divide into separate clusters, and the distance decay of similarity. We swept the drift model's interaction range, between-node mixing rate, and innovation rate, and report the full sweep (see the Supplemental Text) rather than a single setting. To extend the test to the wider valley, we added the southeast-Missouri survey assemblages of Williams (1954). We georeferenced both the Phillips-Ford-Griffin and the Williams assemblages using their Lower Mississippi Valley Survey grid designations against the digitized survey point files, so that the two regions share a single coordinate frame. Decorated classes were harmonized between the two classifications, both rooted in the Phillips-Ford-Griffin framework, into a common vocabulary. We ran the between-region comparison on all decorated classes and on the classes shared by the two classifications (see the Supplemental Text).

The same neutral-drift model is also run forward to produce the phase-like groups seen in Figures 8 and 9. We run it forward in time on the real site geography with no boundary imposed, so any structure in the output comes from geography and time alone. Each site holds a finite population of learners. In each generation, a learner copies the decorated class of another learner, drawn from its own site with high probability and, at a low rate, from another site with a probability that decays with along-river distance. At a low rate, the learner instead adopts a newly invented class, so the repertoire turns over through time, and the assemblages can be seriated. Because the assemblages are not contemporaneous, we sample the simulated field time-transgressively. Each synthetic assemblage is drawn at the time-slice that matches its position on the seriation axis, accumulated over a trailing window to reproduce the record's time-averaging, then subsampled to the observed sherd count. We detect groups in the results using modularity on the Brainerd-Robinson similarity graph, following the same procedure applied to the real assemblages. Because the group labels differ between runs, we summarize Parkin's group as a probability derived from 500 runs. This probability consists of the fraction of runs in which each assemblage falls into the same drift-detected group as Parkin. We report the number of groups, the between-group cultural F_ST, and the co-membership against the named phases, with the full parameter grid in the Supplemental Text.

External cross-check

The transmission test stands on its own. A polity, or any closure of interaction into bounded groups, however, should also leave non-ceramic traces. We assembled three further lines against which to check it. One genuinely independent of the ceramic data (the lithic exchange economy), one that shares its collection substrate (settlement organization), and one that re-examines the same assemblages by a different method (an independent ordination). We assessed settlement organization using the rank-size distribution of mound area within the basin and the distribution of mound height, asking whether a single center dominates as a redistributive apex, the classic archaeological correlate of a ranked society, or whether size is graded (Peebles and Kus 1977). For the political economy of exchange, we drew on Cobb's (2000) regional analysis of the production and distribution of Mill Creek and Dover chert hoes, the capacity-linked industrial goods whose control would mark economic centralization, rather than measuring it ourselves. We also compared our seriation result to Mainfort's (2003) independent nonmetric multidimensional scaling of overlapping late-period assemblages, a published ordination of 39 assemblages from the central valley and adjacent areas that tests whether the named phases form discrete, bounded clusters or a gradational continuum.

Results

The four signatures distinguish bounded groups from drift in principle. However, given the resolution of the LMV's record, only one of them detects the difference. The idealized validation behaves as designed. When interaction is made among closed, bounded groups, all four signatures move together, while drift and the other mimics each move only a characteristic subset. All four are tightly correlated under closure (mean absolute pairwise correlation 0.88) yet nearly independent under the mimics (0.19 to 0.29), so their joint movement is informative rather than redundant (Figure S1). The record-matched recovery is the more informative (Figure 4). When a known group-closure signal is injected at the real configuration and scored after size control, only cultural F_ST tracks it monotonically, climbing from the null toward a strong positive trend as the injected strength rises. The injected strength s ranges from 0 (pure drift) to 1 (strongly bounded, sharply separated groups). The neutral departure is non-monotonic in conformity, the spatial boundary is unresponsive at the three spatial clusters the data support, and seriation coherence responds only weakly. With the false-positive rate held at five percent, F_ST recovers a closure of strength s greater than or equal to 0.5 with power above 0.8, the threshold rising to 0.7 under the record's time-averaging. The other three signatures are not reliable discriminators at this resolution, and we treat them as weak corroboration only. Thus, at this record size and with time-averaging, the one signature that can distinguish a bounded group from drift is between-group variance.

Applied to the assemblages of the St. Francis basin, that one reliable signature shows no closure into bounded groups, and the apparent signals in the others are confounded by sample size (Figure 5). The raw trajectory along the seriation axis seems to dissociate. Two of its trends, however, are sampling artifacts. Because assemblage size trends with seriation position, the size-sensitive estimators move with it. Once all assemblages are trimmed to a common sherd count via subsampling, cultural F_ST falls from +0.60 to +0.01. Its uncertainty range runs from no closure to a moderate signal, so it is uninformative rather than a resolved positive. The neutrality departure is likewise size-sensitive (i.e., its raw -0.94 only weakens to -0.83 under size control). The recovery test shows that it does not track closure in either direction here, so it, too, cannot be interpreted as a transmission signal. On the recovery curve, the size-controlled empirical F_ST corresponds to a nominal injected strength of about 0.35, below the resolution limit. The record shows no detectable closure of moderate or greater strength: such a closure would have registered and is absent, though the negative cannot rule out a weaker one that, given the record's size and time-averaging, is underpowered to resolve. The next positive control shows that the structure actually present is the one produced by neutral drift. Not only is no bounded group detectable, but the structure present is what continuous, spatially structured interaction would produce.

Three further checks, all pointing the same way, add detail. The first follows each decorated class's rise and fall through the sequence and asks whether the changes resemble the undirected fluctuation of unbiased copying or the directed change a deliberate marker would exhibit. Seven of the nine testable classes are consistent with unbiased drift (i.e., the frequency increment test) (Feder et al. 2014). The second compares how the assemblages' overall diversity changes over time against the range that plain drift would produce,

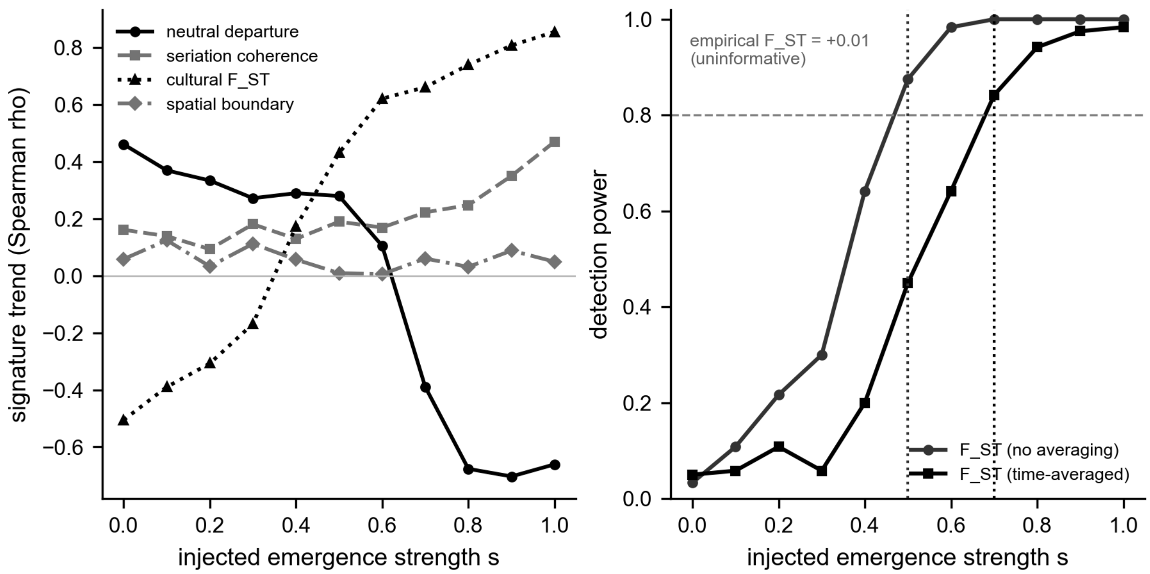

Figure 4. Record-matched, size-controlled signal recovery. Left: each signature's response (Spearman trend versus ordinal position) to a known group-closure signal injected at the real configuration of 29 assemblages and subsampled to a common count. Only cultural F_ST tracks the injected signal monotonically. The neutral departure is non-monotonic in conformity, the spatial boundary is unresponsive at three spatial clusters, and seriation coherence is weak. Right: detection power of the calibrated F_ST test against injected strength, with and without the record's time-averaging (false-positive rate 0.05). F_ST reliably recovers a closure of strength s greater than or equal to 0.5 (0.7 time-averaged). The size-controlled empirical F_ST (+0.01) falls below the resolution limit, its bootstrap interval spanning no closure to a moderate signal.

and finds that it stays within that range (a non-equilibrium neutral test) (Kandler and Shennan 2013). The third asks whether the assemblages grow more different from one another toward contact, as would fragmenting groups. They do not. Neiman's interassemblage distance shows no divergence trend along the sequence (Spearman rho = +0.37, not significant). The typical difference between spatial clusters is barely larger than the difference within them (a ratio of 1.13). Here, the most diverse assemblages are also the most central, the pattern drift produces in a single shared community (Figure S2) (Neiman 1995). The verdict that no bounded-group closure is detectable also holds across the analyst's choices. Across twenty-seven combinations of the basin's spatial cutoff, the number of ordinal bins, and the number of spatial clusters, no closure appears in any of them, and it holds equally under the watershed and corridor basin definitions. Consistent with the recovery experiment, none of these checks shows a resolvable departure from neutral transmission, and we treat them as corroboration of the size-controlled F_ST result rather than as independent signals.

A positive control identifies which of the two processes the basin's structure matches. Rather than rest on a failure to detect, we asked which of the neutral spatial drift and bounded groups reproduces the observed decorated-ceramic structure. Following the finite-population network-drift model of Lipo, DiNapoli, Madsen, and Hunt (2021), we simulated neutral copying on the actual coordinate layout of the basin assemblages, with interactions decaying with distance and time averaging imposed on the record. We compared it against a bounded-groups process on the same geography. By sweeping the interaction range, the between-node mixing rate, and the innovation rate, neutral drift reproduces the observed distance decay, network modularity, cultural F_ST, and distance-controlled boundary excess. The region of parameter space that brackets all four at once is a sub-basin interaction range of about 24 km with low between-node mixing. We report the full sweep rather than a single tuned point. At shorter ranges, drift overproduces distance decay. At higher mixing, it underproduces the structure, so the all-four match holds only in a plausible but specific corner of the space (see the Supplemental Text). The bounded-groups process, by contrast, overshoots the three structural signatures by roughly two to seven times across its range, predicting far sharper community

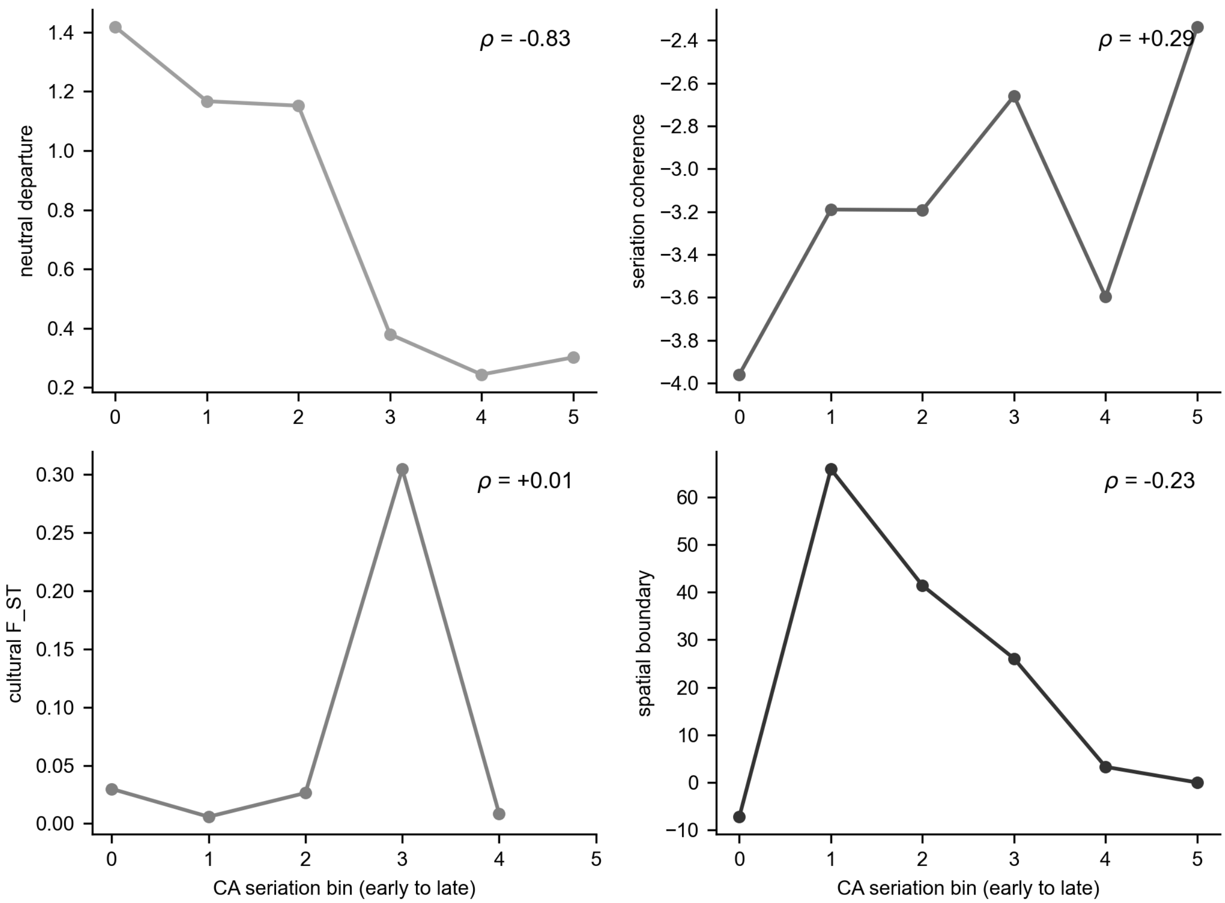

Figure 5. The empirical four-signature trajectory along the CA seriation axis for the St. Francis basin (29 assemblages, six bins), size-controlled by subsampling to a common count. The reliable signature, cultural F_ST, shows no trend (+0.01, down from a raw +0.60 that was a sample-size artifact). The neutral departure, spatial boundary, and seriation coherence are not reliable discriminators at this resolution (Figure 4) and are shown for completeness. Per-panel rank correlations are annotated.

structure than the data demonstrate. Under plausible parameters, the basin's decorated-ceramic structure is entirely reproducible by neutral spatial drift. This result explains the illusion of phases as units: the observed pattern does not require bounded groups, which would have left a sharper mark than is present (Figure S3). The control also bounds the spatial signature itself. A positive distance-controlled boundary excess, which we introduced to separate a sharp social boundary from smooth isolation by distance, is reproduced by stochastic neutral drift on a finite network, so on its own it does not certify a boundary.

The seriation signature does not register closure as a trend at this record's resolution (Figure 4). The structure it returns still directly characterizes Parkin's position. As described above, the deterministic method accepts an ordering only when every decorated class is at once unimodal and continuous, so a set of assemblages drawn from one shared, drifting pool resolves under a single ordering, while learning partitioned among groups breaks into several mutually incompatible orderings (Dunnell 1970; Lipo et al. 2015). The number of maximal co-seriable groups, together with the number of assemblages any one assemblage joins, measures fragmentation directly (Lipo et al. 1997).

Based on the analyses, Parkin is a seriation bridge rather than the anchor of a bounded community. Deterministic seriation of the basin assemblages returns a deeply fragmented, bridge-dominated structure rather than a single coherent ordering. At a continuity threshold of 0.1, it yields 35 maximal co-seriable groups with a maximum size of four, and 17 of the 29 assemblages belong to more than one group (Figure 6). The absolute counts scale with the threshold, but the fragmented, overlapping topology is stable across thresholds. Parkin is one of these bridge assemblages, a member of four groups and ranked eighth of 29, positioned across orderings rather than at the center of one. The empirical pattern is the one Lipo, Madsen, and Dunnell (2015) reported, with Parkin at the intersection of multiple non-co-seriable orderings rather than at the center of a single closed interaction community. They interpreted that bridging position as a possible sign of incipient hierarchy. Based on these analyses, we interpret it as a site that bridges several orderings and serves as a hub of interaction, rather than the seat of a bounded group.

The features of Parkin single it out. However, in comparison, its prominence sits within a graded distribution rather than standing apart as a redistributive apex. Parkin has the basin's tallest platform mound, at 23 feet, ranking first among the 55 basin sites with a recorded mound height (Figure 7). The distribution is graded, however, with no gap separating Parkin from the next centers, the tallest-to-second ratio being 1.0. The settlement record thus shows a prominent center within a graded size distribution, not an administrative tier set above the rest (Hally 1993, 1996), the same picture the transmission test returns (Figure 7).

Discussion

The three lines of evidence converge on a single conclusion. The Parkin phase is not a bounded interaction group, and Parkin is not the capital of a polity. Within the St. Francis basin, Parkin is the largest and most monumentally elaborated center. Yet, its decorated-ceramic record shows no closure of interaction into bounded groups, and the structure present is reproduced by neutral drift across the basin's geography. In contrast, a bounded-groups process would have left a far sharper mark. The settlement and exchange records say the same. Neither the graded mound distribution nor the decentralized chert-hoe economy shows the centralization a polity requires (Cobb 2000). The exchange evidence is regional rather than basin-internal, so we read it as consistent with, not proof of, decentralization at Parkin.

Of these corroborating lines, settlement and exchange are independent of the ceramic data, whereas Mainfort's ordination corroborates the seriation result for the same assemblages using a different method (Mainfort 2003). We are explicit about the scope of the ceramic claim. The transmission analysis bears on whether interaction closes into bounded, like-with-like groups, not on group-level fitness differences or a new unit of selection (Richerson et al. 2016), which assemblage data cannot address. It likewise does not isolate demographic structure, which can shape assemblage diversity independently of any closure of interaction (Henrich 2004; Powell et al. 2009). The phase is built on a single observation, that assemblages resemble their near neighbors more than their distant ones, and that pattern is exactly what spatially structured drift produces on its own. At the scale at which the phase is drawn, it therefore marks continuous interaction shaped by geography rather than a bounded society. Parkin's ceremonial and demographic prominence was real. The polity that prominence is taken to imply is not in the transmission record. An independent precedent for this

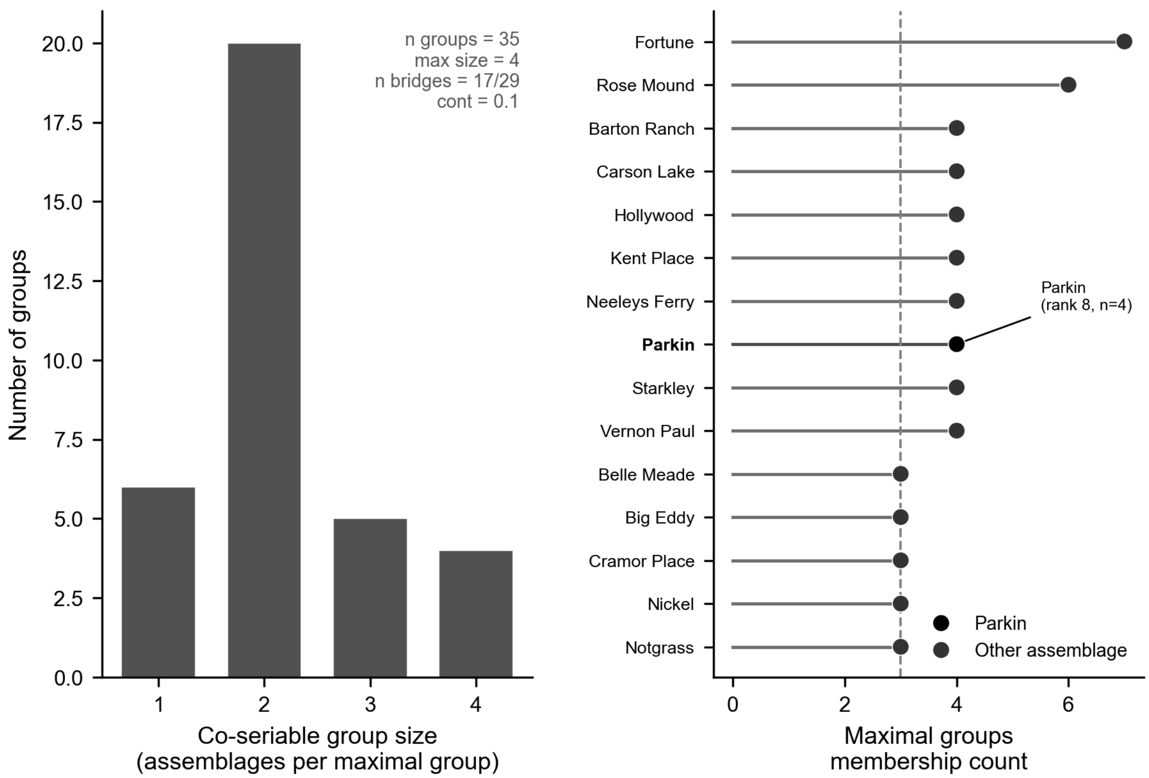

Figure 6. Deterministic frequency-seriation (IDSS) structure of the basin assemblages. Left: distribution of co-seriable group sizes, a deeply fragmented structure (35 maximal groups, maximum size four). Right: the most connected assemblages by group-membership count, with Parkin highlighted as a bridge (member of 4 groups, ranked eighth of 29), positioned across orderings rather than at the center of one bounded community.

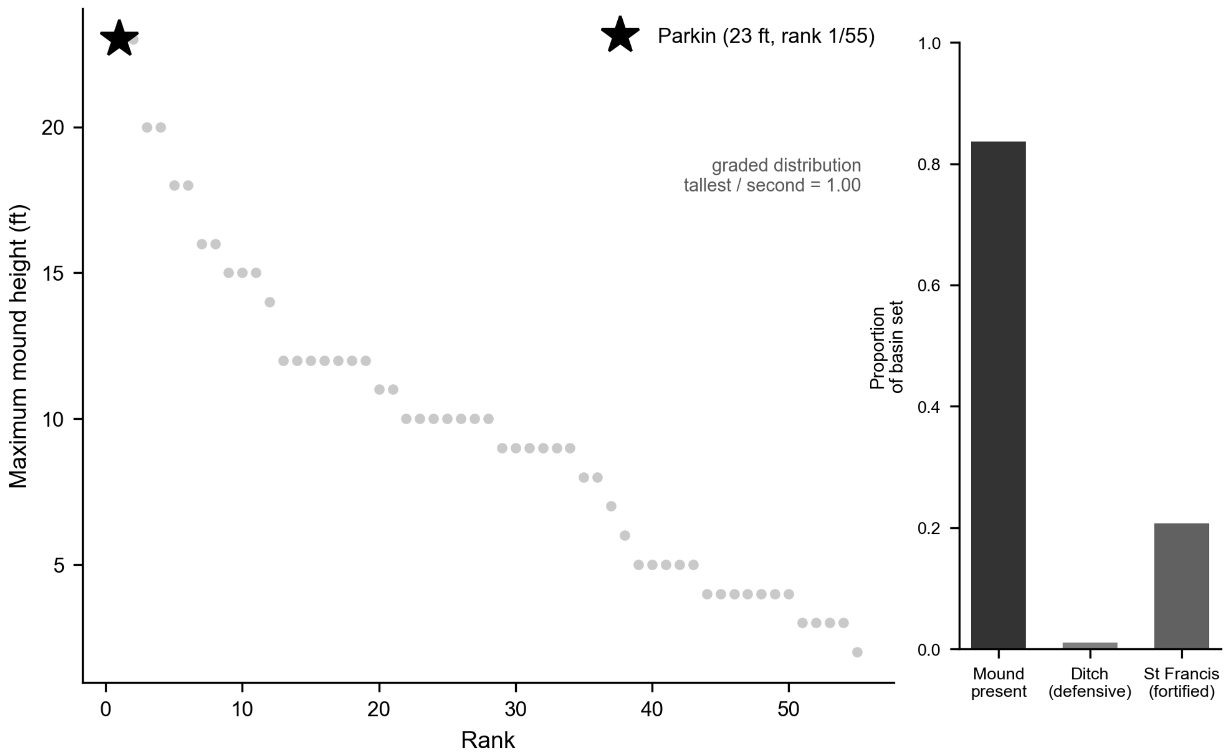

Figure 7. Settlement cross-check. Main panel: within-basin ranking of maximum mound height, the recorded, like-for-like field (Parkin's mound area is under-coded as zero and is not used). Parkin has the tallest mound (23 ft, rank 1 of 55). The height distribution for the mounds is graded, with no gap separating it from the next centers (tallest-to-second ratio 1.0), indicating that Parkin is a prominent center rather than a redistributive apex. Inset: the proportion of sites with a mound, a defensive ditch, and St-Francis-type fortification across the basin set (n = 92).

decoupling appears in the region's own ethnohistory. The best-documented Mississippian paramountcy, the sixteenth-century province of Coosa, spanned two ceramic traditions and site clusters that were separable at the phase level, so what bound them together was political affiliation rather than shared decorated style (Anderson 1996:250–251). A known polity need not coincide with a single ceramic phase, and the converse inference, from one phase to one polity, carries no warrant on its own. This interpretation is not foreign to the Central Valley literature either. In the standard regional synthesis, McNutt (1996:250) characterized the distinctions among the late Mississippian phases as little more than local elaborations, ceramic drift, on a single shared pottery complex.

That characterization can be made constructive rather than merely consistent with the data. Run the finite-population network-drift model forward on the real configuration of the assemblages, with copying that decays with along-waterway distance and no boundary imposed, and the structure that the phase concept interprets as a society reappears on its own (Lipo et al. 2021). Across the 55 decorated assemblages of the wider lower Mississippi Valley, the model repeatedly produces a few spatially coherent, phase-like groups (Figure 8). The deposits are not contemporaneous, which is why they can be seriated at all and why some carry more of the late classes than others, so each synthetic assemblage is drawn time-transgressively at the run-time that matches its seriation position. Holding time fixed to isolate the spatial signal, drift yields two to five groups, the number of which increases with geographic extent. Restoring the staggered deposition ages collapses the count into two time-defined groups because assemblages sampled at different points in the sequence carry different repertoires. Either way, the number of drift-generated groups falls well short of the seven named phases the regional scheme draws across the same assemblages. The between-group cultural F_ST of those groups settles at the observed level rather than the higher value a bounded group would leave. The named phases subdivide the valley more finely than the transmission structure supports. Where culture history draws seven bounded societies, drift on the geography yields at most a handful of coherent groupings, so most of the phase boundaries mark divisions that a continuous interaction field does not carry. Whether any single named phase exceeds this drift expectation is a question we can now pose one phase at a time. That structure, rather than population size, governs how drift partitions diversity in finite populations is well established, and a localized neutral process is easily mistaken for biased transmission when diversity is read at face value (Barton 1996; Premo 2016; Premo and Scholnick 2011; Schneider et al. 2016). The result is not contingent on a tuned parameter set. Across hundreds of runs that spanned a grid of interaction length, between-node mixing, and innovation rate, two or more spatially coherent groups appear in every run, and the between-group F_ST sits at or below the observed value in 85 percent of them. The St. Francis basin that anchors this paper behaves the same way as a positive control (Figures S4 and S5).

The same model lets the named phases be tested one at a time, in the culture historian's own terms. Taking Parkin as the focal case, membership in its group is a probability rather than a fact. Across the 500 runs, each assemblage shares Parkin's drift-detected group some fraction of the time, a graded field that falls off smoothly with river distance from Parkin rather than the bounded set a phase map implies (Figure 9A). The Parkin-phase assemblages do separate from the others by this measure. That separation is what drift predicts, because they lie near Parkin and so co-occur with it more than distant assemblages do (Figure 9B). The discriminating test, which we call the Parkin pull-out, is whether the between-group cultural F_ST across the Parkin-versus-others partition exceeds the drift null. It sits at the extreme upper tail of the null and survives most single-site deletions. The contrast is nonetheless small. The null's own tail extends beyond it, and time-averaging, which deflates between-group variance, is not modeled in the null (Figures 9C and 9D). It is at most a weak hint of structure beyond drift, not a demonstrated social boundary. Geography and time account entirely for the pattern, and the apparent groups are products of a single neutral process.

This finding provides the strongest case for the alternative hypothesis. The most direct argument for an emerging hierarchy holds that Parkin sits on agriculturally poor soil and so must have drawn tribute from the productive villages around it (Phyllis A. Morse 1990). That inference is specific and deserves credit for that, but it is circular. The tribute it concludes is the tribute it assumes, and no independent evidence of tribute flow accompanies it. Parkin's position at the confluence of the St. Francis and Tyronza is also explained by control of movement and communication, as well as by extraction. The transmission and exchange records both show the absence of the centralized control the tribute model requires. A second argument points to skeletal evidence from the lower Mississippi (Yazoo) Delta rather than the St. Francis basin, indicating a

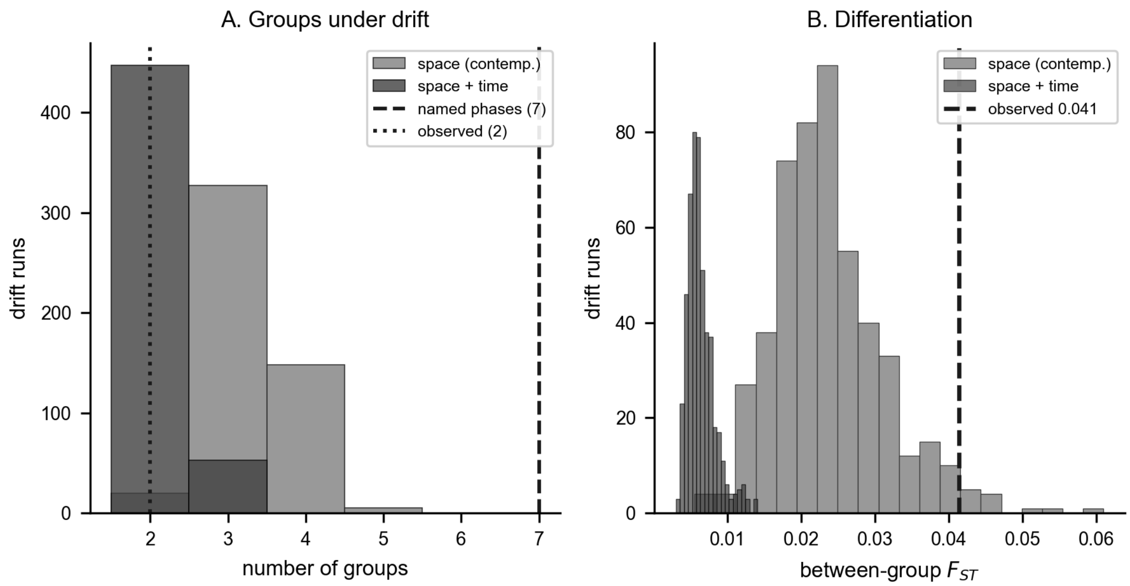

Figure 8. Phase-like groups produced by neutral drift across the wider lower Mississippi Valley, with no bounded group imposed, summarized over 500 realizations. The finite-population network-drift model runs forward on the river-network geography of all 55 Mainfort-PFG decorated assemblages, and groups are detected as greedy-modularity communities on the Brainerd-Robinson similarity graph (Supplemental Text). In the model, each site copies decorated classes primarily from itself and, at a low rate, from other sites, with a probability that decays with river distance, while innovation turns the repertoire over through time; synthetic assemblages are sampled time-transgressively at each site's seriation position and subsampled to match the observed counts. (A) Distribution of the number of groups across the 500 runs, under contemporaneous (pure-space) and time-transgressive sampling, with the seven named culture-historical phases and the observed cluster count marked. Pure spatial drift yields two to five groups, and the count grows with geographic extent; time-transgressive sampling collapses it toward two. Either way, drift produces far fewer groups than the seven named phases, so the phases subdivide the valley more finely than the transmission structure supports. (B) Between-group cultural F_ST across runs against the observed level (0.04); the observed differentiation sits at the upper edge of the contemporaneous-drift distribution, so drift reproduces it. Parameters are the positive-control corner (interaction length 24 river-km, mixing 0.02, innovation 0.012); the basin positive control and the parameter-grid robustness are in Figures S4 and S5.

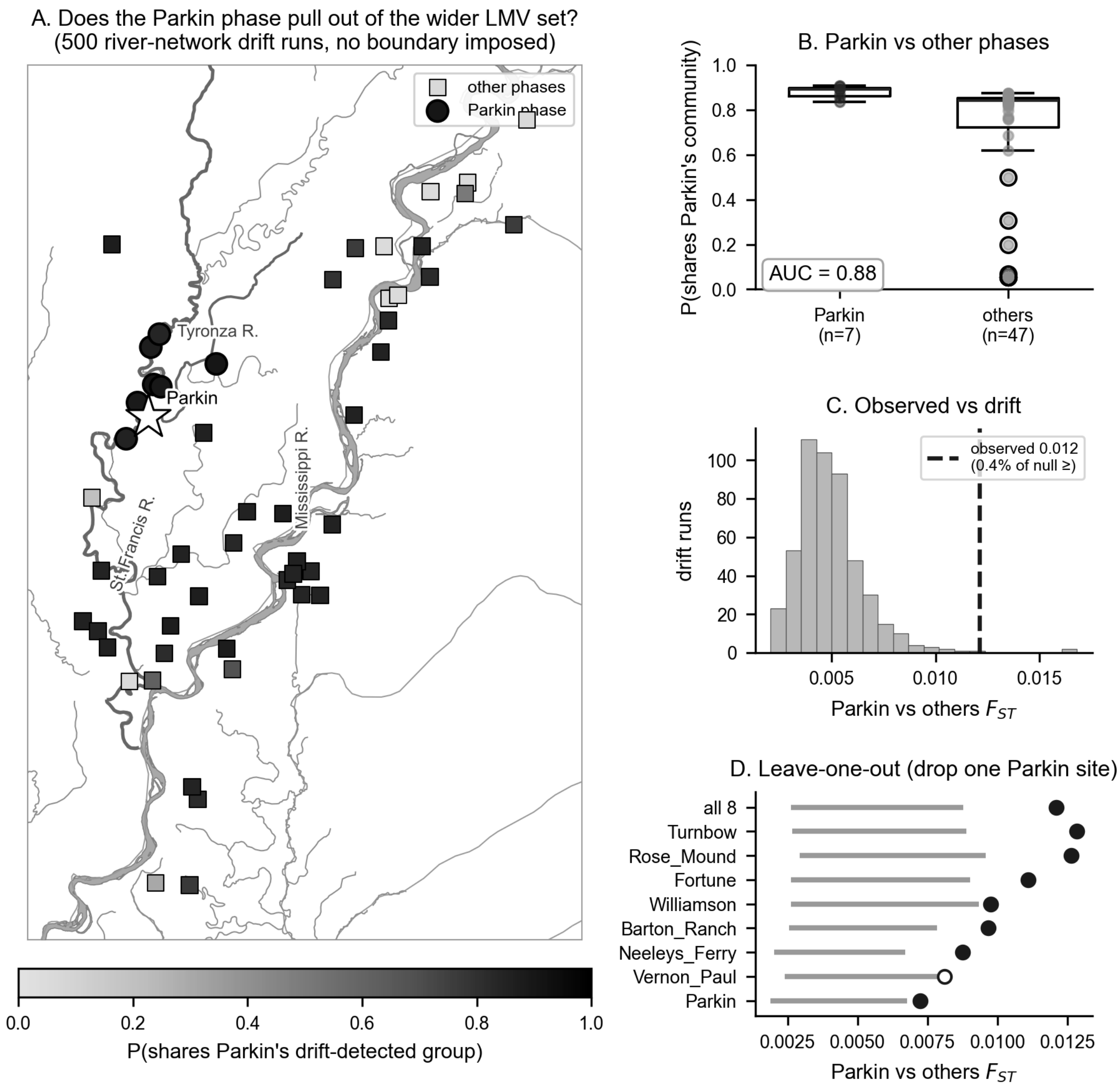

Figure 9. Does the Parkin phase separate from the other named phases by more than drift? All 55 Mainfort-PFG decorated assemblages, labeled by culture-historical phase after Mainfort (1996b), under time-transgressive river-network drift with no boundary imposed (500 realizations). Because group labels differ between runs, Parkin's group is summarized as the fraction of runs in which each assemblage falls in the same drift-detected group as Parkin. (A) Each assemblage is shaded by the probability that it shares Parkin's drift-detected group (star), with Parkin-phase assemblages drawn as circles and other phases as squares; the probability is a graded field that falls with river distance from Parkin. (B) That co-membership for the Parkin phase against the other phases (AUC 0.88) reflects the separation drift predicted by geography. (C) Observed Parkin-versus-others between-group cultural F_ST (dashed) against the drift null; 0.4 percent of runs reach or exceed it, and the null's tail extends past it. (D) Leave-one-out jackknife: the observed F_ST with each Parkin assemblage relabeled into the comparison group, against the 2.5 to 97.5 percentile band of its matched per-drop null. Open points fall inside the band (drift-consistent), filled points outside; seven of the eight single drops stay outside, so the small contrast does not rest on one assemblage.

better diet at a mound center than at a nearby satellite village, consistent with differential access within a hierarchy (Ross-Stallings 2017). The contrast is real but thin: the satellite sample numbers only fourteen individuals, and health differences between the two sites need not arise from an extractive hierarchy. Neither argument supplies the regional, structural signature that the drift-versus-groups comparison seeks but fails to find.

Three limits bound the result. The first is resolution. As the recovery experiment showed (Figure 4), on a record of this size, the discrimination reduces to a single confound-controlled signature, cultural F_ST, and its size-controlled empirical value is uninformative. The negative is informative for a moderate or greater closure, which would have registered. It cannot, however, exclude a weaker one. It is also bounded in form. Recovery succeeds only for closure expressed as between-group divergence, so a closure carried purely by within-group conformity, or by social structure not aligned with the spatial clusters, would lie below detection here at any strength (Supplemental Text). The second limit is the ordinal chronology. The basin radiocarbon corpus pools to five seriation anchors across a calibration plateau, so the axis is a relative order oriented toward later time rather than a calendar rate. This matters less than it might because the discriminating quantity, the size-controlled F_ST trend, is a magnitude that does not depend on the axis orientation, and the settlement and exchange lines do not depend on the ceramic chronology at all. The third limit is time-averaging, which damps between-assemblage variation and reduces apparent turnover (Coco and Iovita 2020; Kidwell and Tomašových 2013; Perreault 2018; Tomašových and Kidwell 2009, 2010; Youngblood et al. 2023). Rather than arguing that this makes the absence conservative, we built the damping directly into the recovery experiment, thereby raising the detection threshold reported above. The verdict is stable across the watershed (19 assemblages) and corridor (29) basin definitions.

This work makes methodological and substantive contributions. The four signatures distinguish a bounded interaction group from drift in principle, and an idealized simulation confirms that closure drives them together while drift and the other mimics do not (Figure S1). The calibration is the part that enables us to apply this approach to real data. It asks, before any similarity structure is interpreted as social, which signatures a given record can support, and how strong a closure each could detect. The Parkin record shows the test reducing to its single most reliable component, cultural F_ST. We are explicit about what that costs. Had F_ST shown a positive trend, it alone could not have separated a real group from environmental patchiness, which also drives between-group variance (Table 1). In the positive case, the generative drift-versus-groups comparison, not F_ST alone, would have had to carry the discrimination. The negative is cleaner because the present structure is positively matched by drift and overshot by bounded groups. This result shows that social structure may be measurable through decorated ceramic assemblages. We can also assess whether the Parkin phase was a polity by examining whether interaction had closed into a bounded group to a degree that the record can resolve. For the Parkin phase, the answer is that it had not, within the bounds supported by the record. The polity the De Soto encountered seems to reflect distinctions drawn in ceremonial practices, fortification, and settlement size, rather than one written into the structure of ceramic transmission.

One result complicates the drift explanation and warrants closer scrutiny. Within the valley, the Parkin phase is the only grouping that hints at structure beyond drift. As the test against the other named phases showed (Figure 9), the contrast persists after removing any single Parkin assemblage. Yet, it stays small enough that the drift null's tail still reaches past it, so geography alone, with no bounded group, remains a sufficient explanation. A center just beginning to differentiate from its neighbors, rather than a finished, bounded group, would leave exactly this kind of faint, not-yet-separable excess. Contact and the protohistoric depopulation would have cut any such sequence short before it could consolidate. Based on the weak evidence, though, we cannot claim this happened. The time-averaged surface record of eight Parkin-phase assemblages does not support the claim, and drift remains the more parsimonious interpretation. The possibility that Parkin was an emerging center nonetheless warrants direct investigation with data that the present record lacks. For example, stratified, excavated assemblages with independent radiocarbon dates are needed to test whether differentiation increases over time rather than remaining flat. Compositional and petrographic sourcing of the ceramics could potentially test whether production and exchange were localized in the center. In addition, the chert-hoe and shell exchange networks (Cobb 2000) and the mortuary and dietary record (Ross-Stallings 2017) could be used to test whether economic and social differentiation track the ceramic hint or run independently of it. Whether Parkin was beginning to pull away from its neighbors, or sits where drift on the river network happens to concentrate interaction, is a question this analysis raises but cannot settle.

The test also extends beyond the St. Francis basin (Figure 10). Applied to the southeast-Missouri assemblages compiled by Williams (1954), the within-region decorated-ceramic structure is again drift-consistent once the earlier Woodland component is set aside. The absence of bounded groups is not peculiar to the St. Francis basin (Figure 11). In both regions, the observed between-cluster F_ST sits at the neutral-drift level and far below the bounded-groups expectation (LMV +0.02, CMV +0.03, against bounded-groups nulls of +0.11 and +0.08). The assemblages intermix rather than separate into clusters. Between the two regions, a genuine discontinuity appears across the roughly 50 km gap separating the central-valley and lower-valley assemblage clusters, with decorated similarity declining by far more than neutral spatial drift would predict for the same geometry. The two regions carry mutually exclusive, distinctive decorated repertoires (Figure S6). This is the scale-dependent pattern the test is designed to detect, with no closure within regions and a sharp difference between them. We report it as a pattern rather than a verdict, because it is confounded with time.

The distinctive central-valley classes are early- to middle-Mississippian, and the distinctive lower-valley classes are late (Phillips 1970; Williams 1954). The step at the boundary cannot be separated from the early-to-late turnover of the repertoire, as reflected in time-averaged surface assemblages. Resolving whether the inter-regional boundary is synchronic and social or a chronological succession requires independent radiocarbon control of the central-valley assemblages.

A second confound compounds the first. The lower-valley assemblages were collected and identified by classes created by Ford (1936), later by Phillips, Ford, and Griffin (1951). In contrast, the central-valley assemblages were collected and classified separately by Williams (1954), so the two sets of assemblages differ in analyst, sorting convention, and class definition as well as in space and time. The shared-class restriction guards against the crudest version of this artifact. A boundary built only from classes unique to each typology would vanish when only the classes both traditions record are compared. The boundary does weaken under that restriction without disappearing. Even the shared classes, however, may have been sorted to subtly different criteria by the two traditions. We therefore cannot rule out that the apparent central-valley and lower-valley boundary is in part an artifact of two separate analytical histories, and that the possibility of continuous interaction across the whole region, with no real boundary anywhere, remains open. That possibility is part of the question this study raises, rather than a nuisance to be set aside, because a boundary in the data is not yet a boundary in the past. The paper's point at the local scale applies with equal force between the regions.

Beyond the Parkin case, the test proposes a standard of measurement for an evolutionary archaeology that has long been clearer about what evolves than about how to measure it (Dunnell 1995a). A signature that interaction has closed into a bounded group should be theoretically grounded, dynamically sufficient, and empirically sufficient, the last meaning matched to the quality of the record rather than to an imagined snapshot. Ours meet the first by construction, each entailed by the transmission model rather than borrowed by analogy (Lipo et al. 2015; Neiman 1995). The third is that we test rather than assume, which is the point of the recovery experiment. Empirical sufficiency is a property a signature has only if it can register a signal at the resolution preserved by the deposit. In the case of the Parkin record, only between-group variance qualifies (Perreault 2019). They meet the second only in part, because the signatures measure states along the seriation sequence rather than the transformation between them. We therefore add two time-domain tests that jointly satisfy dynamic and empirical sufficiency. A generative, time-averaging-aware inference fits a frequency-dependent copying model to the whole sequence, with the record's time-averaging built into the model. It recovers a transmission bias tightly constrained near neutral (95 percent interval -0.02 to +0.06, including zero), which rules out the conformist dynamic a marker would require (Crema et al. 2016). A tempo-and-mode comparison finds the aggregate diversity trajectory best described by neutral drift rather than by a directional, stabilizing, or mean-reverting alternative (Hunt 2006; Hunt et al. 2008). Neither recovers a transition that the static criterion missed, so the negative now holds in the time domain as well as on the static states. The detail is in the Supplemental Text.

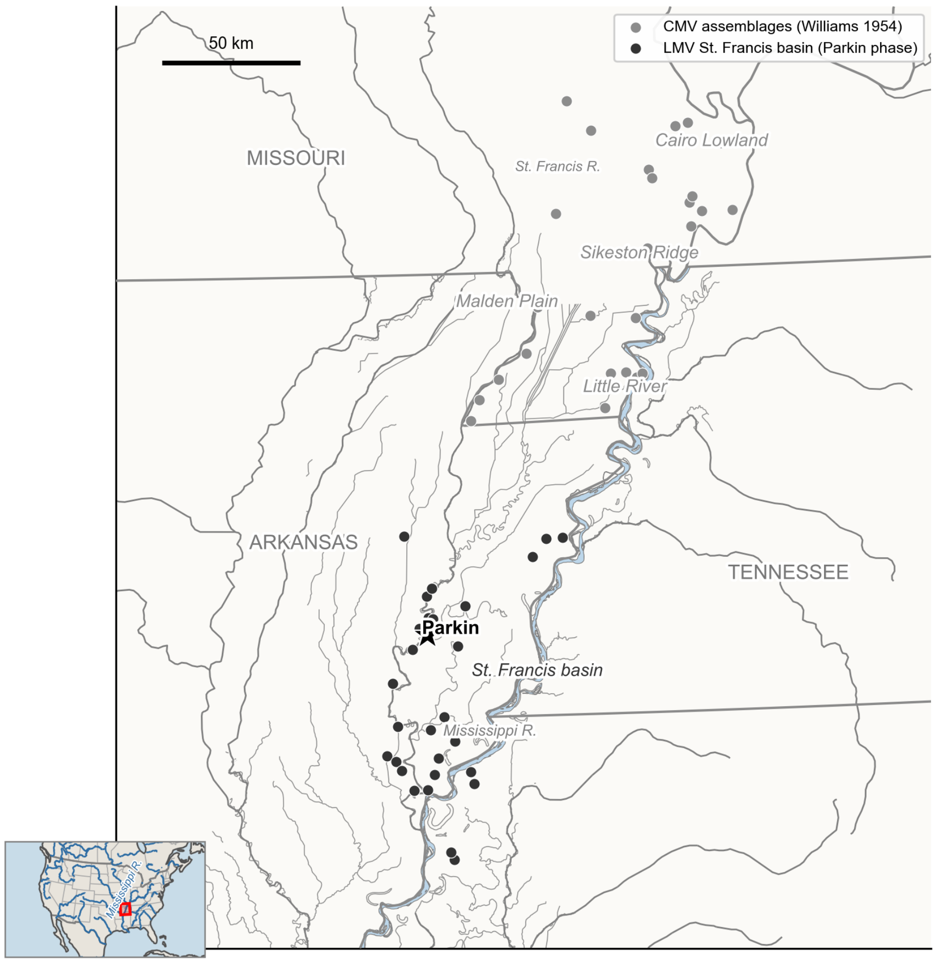

Figure 10. Regional setting of the two-scale comparison. Decorated-ceramic assemblages spanning the central and lower Mississippi Valley: the central-valley southeast-Missouri survey of Williams (1954) (light gray) and the lower-valley St. Francis basin Parkin-phase assemblages (black), identified in the legend, with Parkin starred and a clear spatial gap between the two assemblage clusters. Italic labels mark the named physiographic regions: Cairo Lowland, Sikeston Ridge, Malden Plain, and Little River in the central valley, and the St. Francis basin in the lower valley. Detailed local hydrography is drawn from the Lower Mississippi Survey layer, where it reaches, and from Natural Earth to the north. State boundaries and the North America locator inset are from Natural Earth. The two blocks were collected and classified separately, the central valley by Williams and the lower valley in the Phillips-Ford-Griffin tradition, the distinct analytical histories discussed in the text.

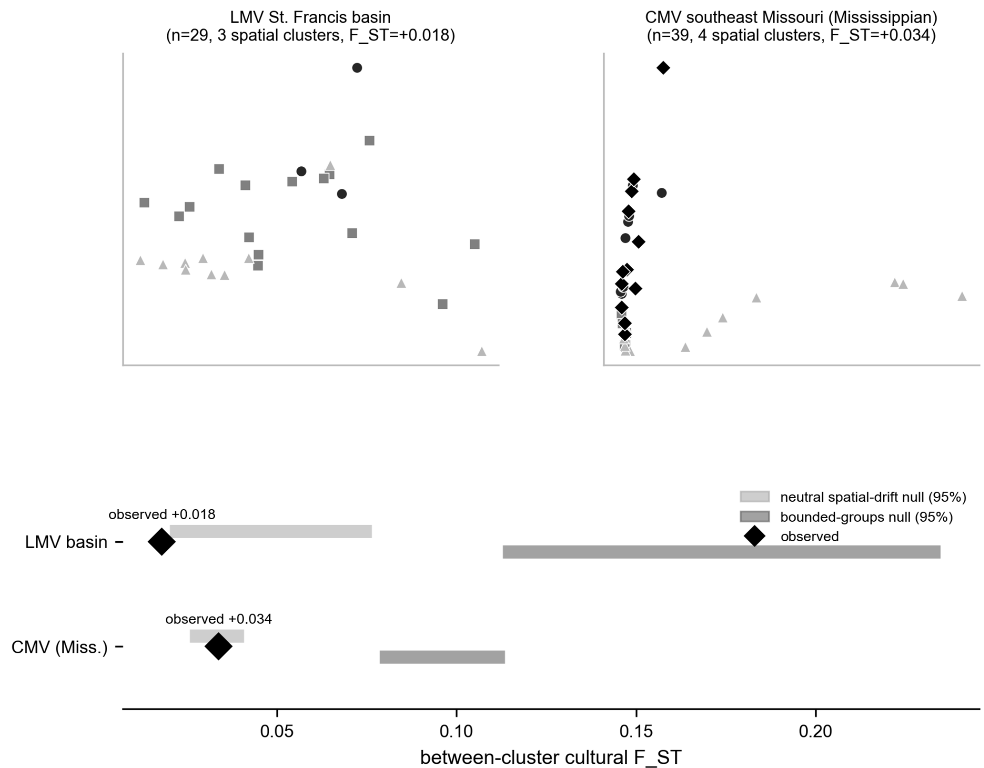

Figure 11. Within-region interaction structure: neither region forms bounded groups. Top: multidimensional scaling of decorated Brainerd-Robinson dissimilarity for the LMV St. Francis basin and for the CMV southeast-Missouri assemblages (Mississippian classes only, so the comparison is not driven by the Woodland-to-Mississippian gradient), points distinguished by shade and marker according to within-region spatial cluster, and the assemblages intermix rather than separating into clusters. Bottom: the observed between-cluster cultural F_ST for each region (diamond) against the 95% nulls of a neutral spatial-drift model (light gray) and a bounded-groups model (dark gray) run on the real coordinates and subsampled to the observed counts. In both regions, the observed F_ST sits at the drift level and far below the bounded-groups expectation, so the within-region structure is drift-consistent at both scales.

Conclusion

The phase is the unit on which central-valley culture history is built, and it carries a silent claim that assemblages grouped by seriation and spatial proximity record a bounded society. For the Parkin phase, that claim does not survive measurement. Parkin is the basin's largest and tallest-mounded center. However, no evidence of settlement, exchange, or the transmission of ceramic decoration indicates that it is the apex of a polity. The greater similarity among nearby assemblages that defines the phase is reproduced by neutral drift across the basin's geography, the ordinary result of continuous interaction structured by site locations. A bounded interaction group would have a sharper structure than the measurements indicate, and Parkin sits across orderings as a transmission bridge rather than at the closed core of one. Extending the test to the wider valley, a decorated-ceramic boundary that exceeds drift appears only between the central-valley and lower-valley blocks. Even there, it is entangled with the early-to-late turnover of the class repertoire. It cannot yet be interpreted as a synchronic social division. Posed in the culture historian's own terms, against the other named phases of the valley, the Parkin phase itself shows at most a weak contrast at the edge of the drift null, robust to single-site removal but small and within the null's reach (Figure 9). The phase, at the scale where it is drawn, is largely indistinguishable from spatially structured drift. That carries a consequence that the discipline should consider. A similarity cluster is not, by itself, a society. Before a phase is interpreted as a social group, the record must be shown to distinguish a real interaction boundary from geographic drift. That calibration, not the phase, provides the basis for an explanation of social structure. When applied to the late pre-contact ceramics of the Mississippi Valley, it shows continuous interaction with prominent nodes rather than a mosaic of bounded polities.

Data and Code Availability

All data and code needed to reproduce the analyses and figures reported here are openly available at https://github.com/clipo/phases. The repository includes the decorated-ceramic class counts, the georeferencing and radiocarbon data, the complete analysis procedure, and instructions for reproducing all results and figures. The code is released under the MIT License, and the data under a Creative Commons Attribution 4.0 International license.

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